



SMART BRIDGE

GRADE 04

BUILDING SMART BRIDGE

MODEL FOR A SUSTAINABLE INFRASTRUCTURE (SDGS)

Theme: Smart Bridge

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Theme-01

Smart Bridge

By the end of the theme Students will be able to:



Use and convert standard units of measurement accurately, and measure bridge components precisely using appropriate tools.



Identify and measure different types of angles (right, acute, obtuse) using protractors, and apply this knowledge to ensure structural integrity and optimal load distribution.



Practice basic programming with Arduino to control sensors and collect data.



Apply the engineering design process to design, build, test, and improve the smart bridge, working collaboratively in teams and developing communication skills.



Combine math skills with engineering and technology in a hands-on project.



Understanding how accuracy is important in building and using technology in real life.

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UNIT 01 [Beginner]

↳ Chapter 01 : Design.

LESSON 01

Smart Bridge.

Time Frame: 20 mins

Objectives: -

- Students will recognize the challenge of safely crossing a river where no permanent structure exists.
- Students will discuss the old wooden bridge, why it failed (flood damage, blocking ships), and learn from past design mistakes.
- Students will brainstorm and suggest innovative ideas, like a **removable or movable bridge**, to meet the current needs.

SEL Relation: -

- **Self-Awareness** learn that it's okay to suggest ideas, even if they need to be improved later.
- **Social Awareness** understanding others' experiences and perspectives can help in problem-solving.
- **Responsible Decision-Making** Reinforces making safe, thoughtful decisions based on facts and input from others.

Challenge Questions of the Lesson: -

- Why was the original wooden bridge not a suitable long-term solution for the village? What design improvements can be made to solve these problems?
- Laila suggested building a strong removable bridge. What STEAM design steps would you follow to turn her idea into a working model?
- How can the new bridge design balance both safety for people and space for boats to pass through the river?

Lesson

1. Engage

Teacher Role:

- Facilitate the storytelling session in an engaging and expressive way.
- Ask guiding and open-ended questions to spark curiosity.
- Connect the story to students' personal experiences or real-world situations.
- Use visual aids, props, or voice modulation to maintain attention.
- Set the stage for the design challenge by highlighting the problem in the story.

Expected From Students:

- Actively listen and respond to the story.
- Share their ideas, feelings, or experiences related to rivers, bridges, or solving problems.
- Ask questions and contribute to group discussions.
- Begin identifying the problem presented in the story.

Parts of the Books Included:

- **STEAM Connection:** Bridge design (Engineering & Design Thinking)

Strategies to use with this part

1. Make Predictions

- **Strategy:** Share the story's title and a related image or drawing.
- **Implementation:**
- Show the title: "*Laila, Adam, and the River Crossing Problem.*"
- Display a simple image of a river with no bridge.
- Ask: "What problem do you think Laila and Adam will face?"
- Record student predictions and revisit them after reading the story

2. Think-Pair-Share

- **Strategy:** Activate prior knowledge and allow idea sharing.
- **Implementation:**
- Ask: "Have you ever had a time when you needed help to solve a problem?"
- Students think silently, then discuss their experience with a partner.
- Invite volunteers to share with the whole class.



3. Picture Prompt

- **Strategy:** Use visual storytelling to spark imagination.
 - **Implementation:**
 - Show a picture of a river, children on one side, and no bridge.
 - Ask: “If you were Laila or Adam, how would you cross this river?”
 - Encourage multiple responses (swim, boat, build something, etc.).
-

4. Role on the Wall

- **Strategy:** Build character understanding.
 - **Implementation:**
 - Draw an outline of a person on a large paper titled “Laila.”
 - Ask: “What might Laila be thinking or feeling during this adventure?”
 - Add student responses around the drawing and refer to it during and after the story.
-

5. Feelings Forecast

- **Strategy:** Connect emotionally with the characters.
 - **Implementation:**
 - Before the story, say: “Today’s story has a big problem in it. How do you think the characters might feel?”
 - Students choose an emotion card or word (excited, scared, curious) and explain why.
 - Revisit these feelings after the story and see if they changed.
-

6. K-W-L Chart

- **Strategy:** Explore what students already know and want to know.
- **Implementation:**
- Create a chart with three columns: **K (Know)**, **W (Want to know)**, **L (Learned)**
- Ask students: “What do we know about bridges or crossing rivers?”
- Fill in the “K” and “W” sections before the story.
- Complete the “L” section after the lesson or activity.

Purpose of Reflection:

Encourages critical thinking and helps students connect story themes to lesson objectives.

Reflection

What caused the need for a new bridge design?

- A) Increased tourism.
- B) Changes in water level.**
- C) new types of ships.
- D) Government regulations.



What problem did Adam mention regarding boats and bridges?

- A) Boats are too heavy to cross bridges.
- B) Boats cannot move under a bridge.**
- C) Bridges are too expensive to build.

Reflection Question Strategies

1. Think-Pair-Share

- **Strategy:** Encourage students to process their thoughts individually, then discuss with a peer.
- **Implementation:**
 - Ask a reflection question: “Why do you think Laila's idea was better than Adam's?”
 - Students think silently for a few seconds.
 - Pair with a partner to discuss their thoughts.
 - Select volunteers to share with the whole class.

2. Hot Seat

- **Strategy:** Bring characters to life by having students answer questions *as if they were* the character.
- **Implementation:**
 - Invite a student to sit in the “hot seat” as Laila or Adam.
 - Ask questions like: “What were you thinking when the grandfather said it would take a long time to cross the river?”
 - Classmates may also ask questions to the character in the seat.

3. Draw & Reflect

- **Strategy:** Use drawing as a tool for reflection and deeper understanding.

- **Implementation:**
 - Ask: “Can you draw what the new bridge might look like based on Laila’s idea?”
 - Below the drawing, students write or discuss: “Why is your bridge design better than the old one?”
 - Share drawings with peers or display them.
-

4. Four Corners

- **Strategy:** Promote physical movement and opinion-sharing around reflection questions.
 - **Implementation:**
 - Label four corners of the room with possible answers or feelings (e.g., “safe,” “unsafe,” “smart idea,” “not sure”).
 - Ask: “Was Adam’s idea of swimming a good solution?”
 - Students go to the corner that represents their opinion and explain why.
-

5. Exit Ticket

- **Strategy:** Encourage individual reflection at the end of the lesson.
 - **Implementation:**
 - Ask a closing reflection question: “What is one thing you learned from Laila’s solution?”
 - Students write a sentence or draw a quick picture on a sticky note or card and hand it in before leaving the session.
-

6. Fishbowl Discussion

- **Strategy:** Promote group reflection and deep conversation.
- **Implementation:**
 - Choose a small group of students to sit in the center (the “fishbowl”) and discuss a question like: “What would you have done differently if you were Adam or Laila?”
 - The rest of the class observes quietly and then shares what they noticed or learned.

Time Frame: 70 mins

Objectives: -

- Master different tools of Tinkercad as mirror tool, duplicate and rotate
- Design a bridge using design software as Tinkercad

Challenge Questions of the Lesson:

- If you couldn't build a removable bridge like Laila suggested, what other creative and safe solution could you design to help people cross the river?
- How can you clearly explain your design idea so that even someone who wasn't part of your group understands how it works and why it's a good solution?

SEL Relation: -

- **Relationship Skills**
Supports communication, teamwork, and listening to others' ideas. And Encourages cooperation, turn-taking, and building on each other's thoughts.
- **Self-Management** Fosters perseverance, focus, and a growth mindset. And Helps students manage frustration, revise ideas, and stay goal-oriented.

Lesson Plan

1. Engage
Teacher Role:

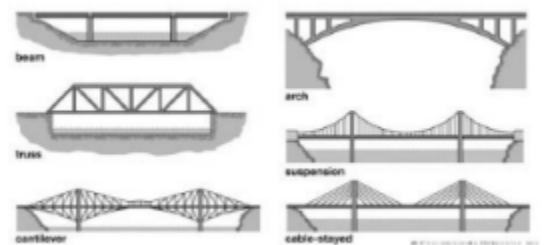
- Facilitate discussion by asking open-ended questions.
- Encourage curiosity and guide students in their online or book-based exploration.
- Introduce different types of bridges using the image.
- Ask follow-up questions to provoke deeper thinking about design and strength.

Expected from Students:

- Explore: Research types of bridges and their real-world uses.
 - Observe: Look closely at the bridge image and compare designs.
 - Reflect: Share their ideas on why some bridges are stronger than others.
 - Predict: Guess which bridge type might work best in different environments.
-

Parts of the Book or Image Included:

- The provided bridge image showing different types (beam, arch, truss, suspension, cantilever, cable-stayed) will be used for visual inquiry.



Strategies to use with this part

1. See, Think, Wonder

- **Purpose:** Encourages students to observe, think critically, and question.
 - **Implementation:**
 - o **Ask:**
 - “What do you see in the bridge designs?”
 - “What do you think the purpose of the different shapes is?”
 - “What do you wonder about how bridges stay strong or carry heavy loads?”
 - o **Have students write or draw their answers.**
-

2. Think-Pair-Share

- **Purpose:** Promotes peer learning and collaboration.
- **Implementation:**

- Ask a question like: “How do you think the shape of a bridge helps make it stronger?”
 - Let students think silently.
 - Then pair with a partner to discuss.
 - Finally, share their ideas with the class.
-

3. Turn and Talk

- **Purpose:** Builds quick discussion around a focused prompt.
 - **Implementation:**
 - Pose a focused question: “Why do you think some bridges have cables while others have arches?”
 - Students turn to the person next to them and discuss for 1–2 minutes.
-

4. Visible Thinking: “Parts, Purposes, Complexities”

- **Purpose:** Helps break down a complex structure (like bridges).
 - **Implementation:**
 - **Guide students to identify:**
 - Parts – “What parts can you see in this bridge?”
 - Purposes – “What is each part doing?”
 - Complexities – “What might be challenging about building this?”
-

5. Concept Mapping

- **Purpose:** Visually organizes thoughts about different bridge types.
 - **Implementation:**
 - **Students draw a simple web chart with the word “Bridge” in the center.**
 - **From each type (e.g., beam, arch, truss), they add:**
 - Strengths
 - Where it might be used
 - Materials it might need
-

6. I Notice – I Wonder

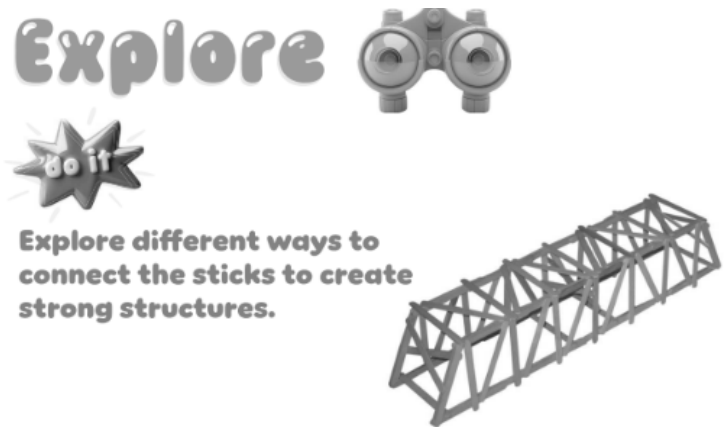
- **Purpose:** Builds observation and curiosity.
- **Implementation:**
 - **Students write two lists:**

- “I Notice...” (about shapes, cables, supports)
 - “I Wonder...” (about materials, stability, or how they’re built)
 - o **Share with class or in small groups.**
-

7. Bridge Detective (Role Play Strategy)

- **Purpose:** Makes the observation task playful and imaginative.
- **Implementation:**
 - o **Students become “bridge detectives” and analyze:**
 - Which bridge would be best for a city with heavy traffic?
 - Which one could be removed or moved like Laila suggested?
 - o **Encourage using evidence from the image to support their detective decision.**

2. Explore



Teacher Role:

- Facilitate safe and creative hands-on exploration.
- Ask guiding questions to provoke thinking and experimentation.
- Encourage collaboration and idea sharing between students.
- Provide all materials, explain safety rules, and manage time.
- Observe how students test and improve their structures and guide without giving answers.

Student Role:

- Work in pairs or small teams to brainstorm and build bridge prototypes.
- Use different techniques to connect materials (glue, tape, string, etc.).
- Test their bridge structures for strength and stability.
- Record observations, take notes, and revise their design based on results.
- Respect lab materials, safety rules, and group collaboration.

Guiding Questions and Possible Answers

1. **Q:** *What makes a structure strong and stable?*
A: The shape (like triangles), how the parts are connected, and the balance of the design.
 2. **Q:** *Which shapes are stronger—squares or triangles? Why?*
A: Triangles are stronger because they don't collapse easily under pressure.
 3. **Q:** *What happens if we connect sticks at different angles or with different materials (tape, glue, string)?*
A: It changes how much weight the structure can hold or how stable it is.
 4. **Q:** *Can we combine more than one shape in a design?*
A: Yes! Combining shapes can make the structure even stronger and more creative.
 5. **Q:** *How can we test if our structure is strong?*
A: By adding weights (coins, small toys) or gently pushing it to see if it wobbles.
-

Step-by-Step Guide: Building a Bridge

Step 1: Set Your Goal

- Decide what your bridge needs to do.
 - Should it hold weight? (Coins, toy cars)
 - Should it be long? (Span between two desks)
 - Should it be **removable**, like Laila's idea?
-

Step 2: Brainstorm Ideas

- Think of **different bridge types** (beam, truss, arch, etc.)
 - Decide which shape you want to try (triangles, arches, layers, etc.)
 - Sketch a **simple plan** of what your bridge will look like
-

Step 3: Gather Materials

Choose from the lab table:

- Craft sticks / skewers / straws
 - Tape, glue, string
 - Scissors, ruler, paper, cardboard
 - Coins or toy cars for testing
-

Step 4: Build Your Structure

- Start with the **base** (make it flat and stable)
 - Add **sides and supports** (use triangle shapes for strength)
 - Reinforce the middle so it can hold weight
 - Use tape or string to connect sticks tightly
-

Step 5: Test the Bridge

- Place your bridge between two objects (like chairs, books, or boxes)
 - Slowly add weight (coins, toy cars)
 - Watch how it holds, bends, or breaks
-

Step 6: Improve Your Design

- Ask: What part was weak? What can we fix?
 - Try adding more supports or changing connection methods
 - Rebuild or reinforce parts that collapsed
-

Step 7: Present and Reflect

- Show your bridge to the class
- Explain your design: "We chose triangles because..."
- Share what worked and what didn't.

Strategies for Implementation

1. Hands-On Discovery Lab

- **Strategy Name:** *Tinker Time*
 - **Implementation:**
 - Provide materials like craft sticks, straws, skewers, tape, string, etc.
 - Challenge students to experiment with different ways to join and shape the materials.
 - Encourage them to make mini structures and test their strength using coins or small weights.
-

2. Design Challenge Cards

- **Strategy Name:** *Mystery Connection Cards*
- **Implementation:**
 - Prepare cards with challenges like:
 - "Build a triangle shape that can hold 5 coins."

- "Connect 3 sticks using only string."
 - Let students pick a card and attempt the task.
-

3. Group Brainstorm and Gallery Share

- **Strategy Name:** *Team Webbing*
 - **Implementation:**
 - Divide students into small groups.
 - Each group fills in a brainstorm web.
 - Groups then take turns walking around to see each other's webs and add "bonus ideas" with sticky notes.
-

4. Think Like an Engineer

- **Strategy Name:** *Engineer's Notebook*
- **Implementation:**
 - Give each student a notebook to sketch designs, list materials they tried, and note which connection method worked best.
 - Encourage reflection: "What worked? What didn't? What will I change?"

3. Explain

Teacher Role:

- Guide discussions about bridge types, their strengths, and real-world use.
 - Highlight key engineering concepts (e.g., force distribution, tension, compression).
 - Connect hands-on experiences with scientific facts and real-world structures.
 - Ask open-ended questions that link the students' bridges to actual bridge examples.
-

Expected from Students:

- Actively participate in discussions and connect ideas from the video and article to their own bridge-building experiences.
- Observe different bridge types and features.
- Share opinions, reflections, and insights.
- Record new information in their notebooks or concept maps.

Watch It

Video: [“How Do Bridges Work?” – Practical Engineering \(YouTube\)](#)



Pre-Watching: Set a Purpose

- **Strategy:** “KWL Chart” (*Know–Want to Know–Learned*)
 - **Implementation:**
 - Ask students what they already know about bridges.
 - Have them write one question they hope the video answers.
 - Example prompt: “What do you want to know about how bridges stay up?”

During Watching: Active Engagement

1. Stop and Jot

- **Implementation:** Pause the video at key moments and ask students to write down one idea, fact, or question they have.
- **Purpose:** Encourages active listening and helps retain new information.

2. Guided Viewing Worksheet

- **Implementation:** Provide students with a worksheet that includes:
 - Keywords to define
 - Spaces to write answers to video questions
 - Diagrams to label
 - **Purpose:** Keeps students on track and gives them something to refer to during discussions.
-

3. Prediction Pause

- **Implementation:** Pause the video before a big reveal or solution, and ask:
 - “What do you think will happen next?”
 - “How would you solve this problem?”
 - **Purpose:** Builds anticipation and activates critical thinking.
-

4. Partner Retell

- **Implementation:** After a video segment, students turn to a partner and summarize what they saw in their own words.
 - **Purpose:** Reinforces understanding through verbal processing.
-

5. Question Catcher

- **Implementation:** Give each student a sticky note or card and tell them:
 - “Write down one question you have while watching.”
 - After the video, collect and sort questions as a class.
 - **Purpose:** Promotes inquiry and sparks group discussion.
-

6. Fact or Myth?

- **Implementation:** Before watching, list common misconceptions or “myths” about the topic.
 - As students watch, they mark which statements are true or false.
 - **Purpose:** Builds curiosity and corrects misunderstandings.
-

7. Video Bingo

- **Implementation:** Create bingo cards with key terms, images, or ideas (e.g., “beam bridge,” “tension,” “triangle,” etc.).
- Students mark off squares as they hear or see them in the video.
- **Purpose:** Gamifies viewing and boosts engagement.

8. Compare and Contrast

- **Implementation:** If watching more than one video or seeing different bridge types:
 - Ask students to make a T-chart or Venn diagram to compare features (e.g., suspension vs. beam bridge).
 - **Purpose:** Supports deeper analysis and categorization.
-

9. Emoji Response

- **Implementation:** Provide a few emoji faces on paper (e.g., 😲 = surprised, 🧠 = learned something new, ❓ = confused).
 - Students hold up an emoji at different points or draw one in their notebook with a short explanation.
 - **Purpose:** Encourages emotional engagement and helps the teacher check for understanding.
-

Post-Watching: Discussion & Reflection

- **Strategy:** “Think-Pair-Share”
 - **Implementation:**
 - Ask: “Which type of bridge do you think is the strongest and why?”
 - Students think silently, pair with a partner to discuss, then share with the class.

Read About It

Article: [Types of Bridges Around the World – YardLink](https://yardlink.com/blog/types-of-bridges-around-the-world)



Read about it

Seven Different Types of Bridges and Why We Build Them - Science and STEAM Team

<https://yardlink.com/blog/types-of-bridges-around-the-world>



Pre-Reading: Activate Thinking

- **Strategy:** *“Bridge Hunt Prediction”*
 - **Implementation:**
 - Show 2–3 photos of bridges.
 - Ask students: “Can you guess what type of bridge this is?” (Arch, suspension, etc.)
 - List guesses and come back to them after reading.

During Reading: Stay Engaged

- **Strategy:** *“Highlight and Code”*
 - **Implementation:**
 - Students read in pairs and highlight key facts:
 - Blue = Bridge type
 - Yellow = What makes it strong
 - Green = Where it's found in the world

Post-Reading: Reflect and Connect

- **Strategy:** “*Bridge Match-Up Chart*”
 - **Implementation:**
 - Create a chart with columns: *Bridge Type*, *Strength Feature*, *Real Example*.
 - Have students fill in the chart from memory, then recheck with the article.
-

By the end of the Explain phase, students should be able to:

- Name and describe at least 3–4 bridge types.
- Explain how their own structure relates to real-world bridge design.
- Understand key design terms like *tension*, *compression*, and *span*.

4. Elaborate & Evaluate

Teacher Role:

- Encourage students to **apply their knowledge** creatively in physical and digital formats.
- Guide students as they **build, test, and modify** their bridge designs.
- Support students using **Tinkercad** by modeling tools and answering technical questions.
- Promote discussion on how **design decisions** affect stability and strength.

Expected from Students:

- Use their understanding of bridge types and strength principles to:
- **Build a physical or digital bridge model**
- **Explain their design choices**
- **Compare** different bridge types

- Reflect on how bridges are designed in the real world to balance **strength, materials, and environment**.

Assessment

Students will:

Complete a short **assessment sheet** (you mentioned attaching it).

Assessment

Focus



Choose the correct answer:

1. What type of triangles do we use in designing our bridge?

- A) Equilateral Triangle.
- B) Isosceles Triangle.
- C) Scalene Triangle.
- D) None of the pervious answers.

Project: Build a Bridge in Tinkercad

Materials Needed:

- Devices with internet access
- Tinkercad accounts
- Rulers (for scale)
- Design sketch (optional)

Step-by-Step Instructions:

You can follow this attachment

<https://docs.google.com/document/d/1DqMme3UwxY4JTUYz522y0pnZG-gyOKPh/edit?tab=t.0>

Challenge (at home)
"Design a Stronger Bridge"

Challenge

Design and build a model truss bridge using everyday materials that can support a certain weight, such as a small toy or a stack of books, while stretching across 1 meter.



- Ask students to:
- Create a bridge using materials at home (paper, cardboard, sticks, tape).
- Or draw a detailed design of a bridge using one type they learned about.
- Encourage them to label the parts and explain:
- "Why did I choose this shape?"
- "How would I make it even stronger?"
- Optional: Test it with small weights and record results with a parent or sibling.

Time Frame: 20 mins

Objectives: -

- Identify the **problem** presented in the story and suggest **possible engineering solutions**.
- Describe the **features of a movable bridge** and explain how it allows both **people and ships** to pass safely.
- Brainstorm and **design a simple movable bridge** model using everyday materials or digital tools.

Challenge Questions of the Lesson:

- If you were one of the students in Miss Sara's class, how would you help Adam and Laila build the bridge?
- Why do you think Laila's idea of a movable bridge is better than the old wooden bridge?

SEL Relation: -

- **Relationship Skills:** They work in teams to collaborate, share materials, and divide tasks fairly during the bridge-building activity.
- **Self-Management:** Students stay focused, manage their time, and persevere through challenges during the designing process.
- **Social Awareness:** They listen to Laila's story and empathize with her desire to help others cross the river safely.

Lesson Plan

1. Engage

Teacher Role:

- Begin by reading or telling the story of Adam and Laila's visit to their grandfather and the need for a movable bridge.
- After telling the story, introduce the problem: Adam and Laila need help in designing a bridge that can move to allow ships to pass while still allowing vehicles and people to cross safely.
- Ask students to think about the main goal of the bridge design and why the bridge needs to be movable.
- Facilitate a class discussion to explore their ideas and reinforce their understanding.
- Provide each student with an EDP paper.
- Walk the students through the "Ask" and "Imagine" steps, where they brainstorm ideas and identify questions to guide their design.

Expected From Students:

- Students will actively listen to the story.
- They will reflect on the problem and begin thinking about potential solutions.
- Students will participate in a discussion to reflect on the story's goals.
- They will share their thoughts on why a movable bridge is needed and what purpose it serves.
- Students will come up with questions they need to ask to design the bridge (Ask).
- They will brainstorm ideas on how to make the bridge movable (Imagine).

Strategies to use with this part



Implementation Strategies:

- **Think-Pair-Share (Reflection):**

- Ask the following questions:

1. **What was the main goal of the bridge design?**
2. **Why did Adam and Laila want to make a movable bridge?**

- **Suggested Answers:**

- **What was the main goal of the bridge design?**

1. To allow ships to pass under the bridge while still enabling people and vehicles to cross.
2. To solve the problem of the bridge being blocked for large ships while maintaining safe passage for everyone.
3. To create a bridge that could be used in both scenarios (for ships and people/vehicles).

- **Why did Adam and Laila want to make a movable bridge?**

1. They needed to allow large ships to pass through the river while also allowing people to cross.
2. A movable bridge could solve the issue of the bridge being too low for ships, while still serving as a passage for other vehicles.
3. They wanted to build a bridge that could function in two ways: to be stationary for vehicles and moveable for ships.

- **Implementation:**

- Students will first think individually about the answers.
- Then, they will pair up to discuss their thoughts.
- Finally, the teacher will facilitate a whole-class discussion where students share their answers.

- **Role Play (Reflection Strategy):**

- After the Think-Pair-Share, the teacher can ask a few students to role-play Adam and Laila explaining their bridge problem to the class, encouraging them to use some of the suggested answers from the class discussion.

Strategies to use with this



let's Think

Use the EDP paper to brainstorm ideas for creating solutions.

Name: _____

Project: _____

Ask

→

Imagine

Implementation Strategies:

- **Ask (Stage 1 of EDP):**
 - Have students think about the types of questions they need to answer in order to design a movable bridge.
 - **Possible Questions:**
 1. How will the bridge move?
 2. What materials do we need to make the bridge?
 3. What kind of mechanism can we use to make the bridge open and close?
 4. How much weight will the bridge need to support?
 5. How wide and long should the bridge be?
 6. How will we ensure the bridge can handle both vehicles and ships?
- **Imagine (Stage 2 of EDP):**
 - Students will brainstorm ideas on how to design the movable bridge.
 - **Possible Ideas:**
 1. A **drawbridge** that lifts up vertically to allow ships to pass.
 2. A **swing bridge** that rotates horizontally to create space for ships.
 3. A **bascule bridge** that tilts to let ships pass underneath.
 4. A **sliding bridge** that moves sideways along tracks.
 5. A **pontoon bridge** that floats and can be adjusted to rise or lower.
 6. Using **hydraulic** or **electric motors** to control the movement of the bridge.
 - **Implementation:**
 - Students will write their questions and ideas on the EDP paper.
 - The teacher will encourage students to be creative and think of different solutions before moving to the next stage in the design process.

Implementation Strategy:

- **Exit Ticket:**
 - Ask each student to write down:
 1. **One question** they think is important for designing a movable bridge.
 2. **One idea** for how the bridge could move.
-

Materials Needed:

- Story of Laila and Adam (printed or told orally)
- EDP Paper (for students to fill out during the Think part)
- Pencils, erasers
- Whiteboard/Marker for class discussions

Time Frame: 70 mins

Objectives: -

- Understand the definition and principles of hydraulic.
- Recognize basics of hydraulic system (pumps, cylinders, valves).
- Understand the properties of liquids under pressure.

Challenge questions of the lesson: -

- How can we control the movement of the bridge using pressure?
- Can you design a simple system to lift a heavy object using pressure? What materials would you use?
- How do the balloon and sponge experiments help us understand how pressure works?

SEL Relation: -

- **Curiosity & Growth Mindset:** They explored the lab and discovered how pressure works with balloons and sponges.
- **Problem-Solving & Creativity:** They used what they found to start thinking of new solutions.
- **Perseverance:** Even when they didn't know what pressure was, they didn't give up—they explored and experimented.

▪ Engage

• Teacher Role:

- Facilitate a discussion based on a relatable, imaginative story.
 - Guide students to make connections between the story and real-world science concepts.
 - Ask open-ended questions that promote curiosity and critical thinking.
 - Observe students' prior knowledge about pressure and movement.
 - Introduce basic vocabulary (pressure, hydraulic, liquid, force) during the discussion.
-

• Expected From Students:

- Actively listen to the story and respond to questions.
 - Share their own ideas about how bridges work and how things can move.
 - Participate in discussion by connecting story events to real-life experiences.
 - Express curiosity about the concept of pressure and how it can help move a bridge.
 - Observe and reflect on simple phenomena (e.g., balloon inflating, sponge squeezing).
-

• Parts of the Books Included:

Strategies to use with this part

Adam
I think we need to build a strong bridge..

Lila replied,
Yes, but how are we going to make it move ?

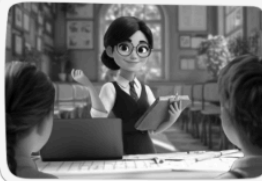
Adam suggested, "Let's ask Mrs. Sara for help.



✓ Understand the properties of liquids under pressure.



Mrs. Sara
Okay kids, why don't you use pressure to make your bridge move?
Both Adam and Lila asked, What is pressure ?



Adam and Lila decided to find out more about pressure. They looked around the classroom and saw how a balloon fills up when you blow air into it, or how a sponge can be squeezed.

They thought maybe pressure could help their bridge move. They were excited to try different ideas and see how pressure could be used in their bridge design.

- **Intro Story:**

Engaging Strategies Used:

Strategy Name: "Storytelling with Role Play"

- **Implementation:** The teacher narrates a mini-story with characters the students know (Adam, Laila, and Mrs. Sara). The teacher uses expressive voice tones and simple props or hand gestures to act out each character. The teacher pauses at key points to let students predict or comment.

Role-Playing (Teacher-in-Role)

- **How to Implement:**
 - The teacher takes on different roles (Adam, Laila, Mrs. Sara) using voice changes, simple props (e.g., glasses for Mrs. Sara), or hand gestures.
 - Invite students to act out parts of the story with you.
 - **Why It Works:** Makes the story come alive and helps students connect emotionally with the characters.
-

2. Story Map Drawing (Visual Narrative)

- **How to Implement:**
 - As you narrate, draw or display a simple map or sketch of the story on the board or chart paper (e.g., river, bridge, lab, balloon).
 - Ask students to help you complete it as the story progresses.
 - **Why It Works:** Supports visual learners and builds engagement through interaction.
-

3. Sound Effects & Props

- **How to Implement:**
 - Use real or improvised sound effects (e.g., "whoosh" for blowing up a balloon, "pop" for pressure release).

- Use props like a sponge, balloon, or mini bridge to demonstrate key story elements.
 - **Why It Works:** Appeals to auditory and kinesthetic learners and makes abstract ideas like "pressure" more concrete.
-

4. Interactive Pause and Predict

- **How to Implement:**
 - Pause at key points in the story (e.g., "What do you think Adam and Laila will do next?") and let students make predictions.
 - Write a few predictions on the board before continuing.
 - **Why It Works:** Builds suspense and encourages critical thinking.
-

5. Freeze Frames (Tableau)

- **How to Implement:**
 - Divide the class into small groups and assign a part of the story to each.
 - They create a "frozen picture" or pose that represents that part of the story.
 - Walk through each scene and have the class guess what's happening.
 - **Why It Works:** Involves students physically, deepens understanding, and promotes teamwork.
-

6. Guided Story Retelling

- **How to Implement:**
 - After telling the story once, involve students in retelling it together.
 - Use prompts like "Who remembers what Adam said?" or "Can someone tell us what Laila noticed at the river?"
- **Why It Works:** Reinforces comprehension and vocabulary.

Q1: What do you think pressure is?

Possible Student Answers:

- “It’s when you push on something.”
 - “Like when I press my hand on the table.”
 - “It’s a force that makes something move.”
 - “When I blow air into a balloon, it grows because of pressure.”
 - “Squeezing something to make it change shape.”
 - “Maybe it’s how we make things go up or down.”
 - “When we push air or water into something and it gets tight.”
-

Q2: What do you think will help you make pressure to move something?

Possible Student Answers:

- “Blowing air into a balloon.”
 - “Using a pump to push air or water.”
 - “Squeezing a bottle or sponge.”
 - “Pushing down a syringe with water inside.”
 - “Using a machine to push liquid.”
 - “Putting water in a tube and pressing it.”
 - “Maybe a balloon can push something if we fill it.”
 - “Using liquid or air to push parts of the bridge.”
-

Teaching Strategy: "Think–Draw–Share"

Implementation:

1. **Think:** Ask students to quietly think about what pressure might mean and how it can help.
2. **Draw:** Have them sketch something that shows pressure in action (e.g., balloon inflating, sponge being squeezed).
3. **Share:** Invite students to show and explain their drawings to the class or in pairs.
→ This helps visual and kinesthetic learners participate meaningfully.

▪ Explore

Teacher Role:

- Provide clear instructions and safety reminders.
- Demonstrate the experiment first, if needed.
- Supervise students as they perform the activity in pairs or small groups.
- Guide students to record their observations.
- Ask probing questions to help students reflect on what they discovered.

• Expected From Students:

- Actively participate in the hands-on experiment.
- Handle the syringes carefully.
- Observe and record the differences between air and water under pressure.
- Compare results and share findings with the class.
- Discuss what these observations tell them about pressure and hydraulic systems.

Activity – Do It: "Air vs. Water in a Syringe"

Materials Needed (per group):

- 2 large plastic syringes (same size, e.g., 20 ml)
 - Cup of water
 - Tissue or small towel (in case of spills)
 - Worksheet or notebook to record observations
-

Step-by-Step Instructions:

1. Take the first syringe and fill it with air.
(Just pull the plunger all the way out—do not put any water inside.)
 2. Place your finger over the nozzle tightly. Try pushing the plunger.
Observe what happens.
 3. Now take the second syringe and fill it completely with water.
(Make sure no air bubbles remain. You can pull and push slightly to remove them.)
 4. Again, place your finger over the nozzle tightly. Try pushing the plunger.
Observe what happens.
-

Reflection Questions & Sample Answers:**1-Fill the syringe with air and push. What do you observe?****Possible Answers:**

- "The plunger moves in, but it's bouncy."
 - "Air is easy to press, and I can still push more."
 - "It felt soft and not strong."
 - "The air got smaller – it compressed."
 - "Some air escaped from the nozzle even when I held it."
-

2-Fill the syringe with water and push. What do you observe?**Possible Answers:**

- "The plunger hardly moves."
 - "It was very hard to push."
 - "The water didn't compress."
 - "It felt tight and strong."
 - "No water escaped if my finger sealed it well."
-

Key Concept Discovered:

- Air (gas) can be compressed under pressure.
- Water (liquid) cannot be compressed and instead transfers the pressure.
- This is why hydraulic systems use liquids, not gases—they transfer force better!

- **Explain**

Teacher Role:

- Guide a discussion to connect the experiment to real scientific concepts.
 - Introduce key vocabulary (e.g., pressure, compress, hydraulic, incompressible).
 - Set a clear purpose before watching the video.
 - Pause during the video for quick checks or questions.
 - Lead post-video reflections, encouraging students to relate it to their hands-on experience.
-

Expected From Students:

- Share observations from the syringe experiment.
 - Use their own words to describe what they think pressure is.
 - Watch the video attentively and respond to guided questions.
 - Connect the video to their experiment and real-life examples.
 - Ask and answer questions using new vocabulary.
-

Parts of the Books Included:

- Any page that defines pressure and introduces hydraulic systems.
 - Diagrams showing how liquids transfer force in a closed system.
 - Visuals or summaries of real-life hydraulic machines (e.g., car lifts, bridges).
 - Vocabulary definitions and key concept summaries.
-

Watch It: <https://youtu.be/28grZdfexT8?si=mXKC68plnHcbRaVg>

Pre-Watching: Set a Purpose

Teacher says:

"Now that we've seen what happens when we use air and water in a syringe, let's watch a video to understand how water helps machines move heavy things!

Watch carefully to answer this question at the end:

Why do we use water in hydraulic systems, not air?"

Strategy: "Question Before Viewing"

- Post the main question on the board.
 - Ask students to predict the answer before watching.
-

During Watching: Active Engagement

Teacher prompts:

- Pause the video and ask:
 - “What do you notice about the way the syringes are connected?”
 - “Why do you think the second syringe moves when the first one is pressed?”

Strategy: "Stop and Think"

- Pause at key moments for quick questions.
 - Allow students to think-pair-share answers.
-

Post-Watching: Discussion & Reflection

Discussion Questions & Suggested Student Answers:

Q: What is pressure in your own words?

A: "It's a push that happens when you press something like water or air."

A: "It's when you squeeze a liquid and the force moves."

Q: Why does water work better than air in a hydraulic system?

A: "Because water doesn't get smaller when we push it."

A: "It can move the pressure better to the other part."

A: "Air just squishes, but water moves the other syringe."

Q: What did you learn about how a hydraulic bridge might work?

A: "It might use water to push parts up and down."

A: "If we use syringes and tubes, we can make something move like a real machine."

Strategy: "Think-Pair-Share"

- Let students answer questions individually.
- Pair up to compare answers.
- Share as a group to build deeper understanding.

4. Elaborate

Teacher Role:

- Encourage creative thinking and hands-on application of learned concepts.
 - Guide students in creating and experimenting with simple hydraulic models.
 - Ask questions to deepen understanding of force and motion.
 - Facilitate group work and encourage reflection after activities.
-

Expected From Students:

- Apply their knowledge of pressure and hydraulics.
 - Build and test models using syringes and water.
 - Observe and describe how changes affect force and movement.
 - Answer review and reflection questions thoughtfully.
 - Complete a take-home challenge to reinforce learning.
-

Parts of the Books Included:

Assessment: Focus

Assessment

Focus

Which of the following is an example of a pneumatic system?

- A) Water hose.
- B) Bicycle pump.
- C) Car engine.
- D) Fish tank.

What is the main difference between hydraulic and pneumatic systems?

- A) One uses solids, the other uses liquids.
- B) One uses liquids, the other uses gases.
- C) One uses heat, the other uses cold.
- D) One uses electricity, the other uses magnets.



Strategies to Apply While Answering:

1. Think–Pair–Share

Purpose: To encourage every student to process and verbalize their understanding before sharing it publicly.

Implementation:

1. **Think:** Ask a question like: “Which is an example of a pneumatic system?” and give students 30 seconds to think quietly.
2. **Pair:** Students turn to a shoulder partner to compare answers and reasoning.
3. **Share:** Call on several pairs to share their discussion with the class.

Example:

- After reading: “*What is the main difference between hydraulic and pneumatic systems?*”, students silently pick their answer, discuss in pairs, then share their logic.

2. Elimination Strategy (Process of Elimination)

Purpose: To develop critical thinking and help students tackle multiple-choice questions strategically.

Implementation:

1. Present a multiple-choice question.
2. Ask students to **eliminate** answers they know are wrong and **explain why**.
3. Focus on identifying keywords (e.g., “liquids” vs. “gases”) and rejecting choices like “electricity” or “solids” that don’t belong.

Example:

- For the question: *“What is the main difference between hydraulic and pneumatic systems?”*
 - Students say: “Let’s cross out A because neither system uses solids. D is not related at all. That leaves B and C.”
-

3. Prediction Before Experiment

Purpose: To activate prior knowledge and promote inquiry before the hands-on practice.

Implementation:

1. Before doing the syringe experiment, ask:
 - “Which syringe do you think will move more – the small or the big one?”
2. Have students **record their predictions** in a science notebook or on the board.
3. After the activity, students reflect on whether their prediction was correct.

Example:

- “I think the small syringe will move the big one more because I have to push harder!”
-

4. Claim–Evidence–Reasoning (C-E-R)

Purpose: To structure scientific thinking and explanations.

Implementation:

1. After the experiment, ask students to make a **claim** based on their observation.
2. Then, they must support it with **evidence** from the experiment.
3. Finally, write a **reasoning** that connects the evidence to the claim using scientific vocabulary.

Example:

- **Claim:** "The big syringe moves the small syringe more easily."
 - **Evidence:** "When I pushed the big one, the small one went farther."
 - **Reasoning:** "This is because the larger plunger spreads the pressure over a bigger area, giving more power to move the smaller one."
-

5. Graphic Organizer – Comparison Chart

Purpose: To visually compare and contrast key concepts (hydraulic vs. pneumatic).

Implementation:

1. Use a Venn diagram or T-chart.
2. Ask students to list features of hydraulic systems (uses liquids, incompressible, strong force) and pneumatic systems (uses gases, compressible, quick movement).
3. Add pictures or real-world examples under each.

Example:

Hydraulic Systems	Pneumatic Systems
Use liquids	Use gases
Lifting bridges	Dentist tools
Strong force	Quick movement

Practice Activity: Using Syringes of Different Sizes

Materials Needed (per group):

- Two syringes of **different sizes** (e.g., 10 ml and 20 ml)
 - A piece of **plastic tubing** that connects both syringes
 - A cup of water
 - Towel (in case of spills)
-

Step-by-Step Instructions:

1. Fill both syringes **completely with water** and connect them using the tube.
 2. Hold both syringes horizontally and **press the smaller syringe** plunger slowly.
 3. Observe how the plunger in the larger syringe moves.
 4. Repeat the process but now press the **larger syringe** and observe the smaller one.
 5. Record what you notice about how much each plunger moves and how much force is needed.
-

Reflection Questions & Sample Answers:

What do you notice when you push the small syringe?

- "The big one moves, but I have to press harder."
- "The plunger in the large syringe doesn't move much, but it feels strong."

What happens when you push the big syringe?

- "The small one moves more and faster."
- "It's easier to push the big syringe, and the small one goes far."

What does this tell you about force and movement?

- "Small syringes need more force to push big ones."
 - "Big syringes can move small ones more easily and with more distance."
 - "Force and movement change depending on the size of the syringes—this is how machines can lift heavy things."
-

Challenge: At-Home Application

Create your own mini hydraulic machine at home using:

- 2 syringes
- Plastic tubing
- Colored water (optional)

Try to design:

- A simple moving arm
- A lifting mechanism
- Or anything that moves using water pressure

Draw or take a picture of your model and write a sentence: *What did you build and how does pressure make it move?*

Here are the **expected answers** to fill in the observation sections based on the activity shown in the image:



With both syringes connected, push the plunger of the large syringe slowly. Observe what happens to the small syringe.

Write your observation:

Note how far the small syringe moves when you push the large syringe. Measure and record the distance travelled by the plunger of the small syringe.

Write your observation:

Now, push the plunger of the small syringe and observe the movement of the large syringe. Record the distance it travels.

Write your observation:

1. With both syringes connected, push the plunger of the large syringe slowly. Observe what happens to the small syringe.

Write your observation:

The plunger of the small syringe starts to move outward slowly as I push the large syringe. The liquid inside transfers the force to the small syringe.

2. Note how far the small syringe moves when you push the large syringe. Measure and record the distance travelled by the plunger of the small syringe.

Write your observation:

When I pushed the large syringe 4 cm, the small syringe moved about 6 cm. The small syringe moved more because the pressure transferred through the liquid made it move faster and farther.

3. Now, push the plunger of the small syringe and observe the movement of the large syringe. Record the distance it travels.

Write your observation:

When I pushed the small syringe 6 cm, the large syringe only moved about 3 cm. It was harder to push because the large syringe needs more force due to its size.

These observations help students understand the **relationship between force, area, and movement** in a **hydraulic system**—a key foundation in pressure and fluid mechanics.

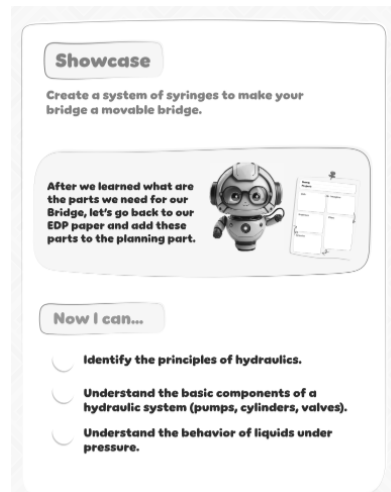
5. Evaluate:

- **Teacher Role: [Facilitate assessments and provide feedback.]**
 - The "Showcase" section, which instructs students to "Evaluate the students' work in the main project and add a rubric so teachers can follow," directly aligns with the teacher's role of facilitating assessments and providing feedback. The request to add a rubric is a concrete step towards enabling teachers to systematically evaluate student work.
- **Expected From Students: [Answer questions, demonstrate understanding, make projects.]**
 - The "Showcase" activity implies that students have completed a "main project" which involves creating a system of syringes to make a bridge a movable bridge. This directly addresses the expectation of students making projects.
 - The "Now I can..." section, which includes statements like "Identify the principles of hydraulics," "Understand the basic components of a hydraulic system (pumps, cylinders, valves)," and "Understand the behavior of liquids under pressure," suggests that students are expected to demonstrate their understanding of these concepts. While the image doesn't show specific questions, the "Now I can..."

section likely serves as a self-assessment or a guide for teachers to ask relevant questions to gauge student comprehension.

- **Parts of the books included:**

- o **Showcase:** This section explicitly focuses on the evaluation of the main project (creating a movable bridge using syringes).
- o **After we learned what are the parts we need for our Bridge, let's go back to our EDP paper and add these parts to the planning part:** This section indicates a prior activity involving an "EDP paper" (likely referring to the Engineering Design Process) where students planned their bridge. The current step involves integrating the knowledge of necessary parts into that plan.
- o **Now I can...:** This section serves as a self-assessment or learning objectives checklist, focusing on the student's understanding of hydraulic principles and systems.



- **Showcase: Evaluate the students' work in the main project and add a rubric so teachers can follow.**

To evaluate the students' work on the "system of syringes to make your bridge a movable bridge" project, a rubric is essential. Here's a sample rubric that teachers can adapt:

Movable Bridge Project Rubric

Criteria	Excellent (4 points)	Proficient (3 points)	Developing (2 points)	Needs Improvement (1 point)
Functionality	The bridge moves smoothly and reliably using	The bridge moves, but may have some inconsistencies or require	The bridge moves with significant difficulty or	The bridge does not move using the syringe system.

	the syringe system.	multiple attempts.	only moves partially.	
Design & Construction	The design is well-thought-out, sturdy, and effectively utilizes the syringe system.	The design is functional but may have some minor flaws in construction or stability.	The design shows some understanding of the concept but may be flimsy or poorly executed.	The design is unclear, unstable, or does not effectively integrate the syringe system.
Application of Hydraulics	Clearly demonstrates an understanding of hydraulic principles in the design and operation.	Shows a basic understanding of hydraulic principles in the design and operation.	Demonstrates limited understanding of hydraulic principles.	Shows little to no understanding of hydraulic principles.
Problem Solving	Effectively addresses challenges encountered during the design and construction process.	Identifies and attempts to address challenges during the process.	Struggles to overcome challenges or does not adequately address them.	Does not identify or attempt to address challenges.
Presentation & Explanation	Clearly explains the design choices, how the hydraulic system works, and any challenges faced.	Explains the design and function of the hydraulic system.	Provides a limited explanation of the design and function.	Unable to adequately explain the design or function.

Scoring:

- 17-20 points: Excellent understanding and execution.
- 13-16 points: Good understanding and execution.
- 9-12 points: Developing understanding and execution.
- 5-8 points: Needs significant improvement.
- **Relation to Standards:** [Link to relevant educational standards.]

To link this project to relevant educational standards, the teacher would need to identify the specific science, technology, engineering, and mathematics (STEM) standards relevant to:

- o **Simple Machines:** Bridges as structures, levers (potentially in the syringe mechanism).
- o **Fluids and Pressure:** Hydraulic systems, Pascal's Law (implicitly).
- o **Engineering Design Process:** Planning, building, testing, and evaluating solutions.
- o **Scientific Inquiry:** Observation, data collection (if applicable, e.g., measuring movement).

Example Connections to Potential Standards (Illustrative - Specific standards vary by region/curriculum):

- o **Next Generation Science Standards (NGSS):**
 - **MS-ETS1-1:** Define the criteria and constraints of a design problem with sufficient precision to ensure a successful solution, taking into account relevant scientific principles and potential impacts on people and the natural environment that may limit possible solutions. (Relates to the design and constraints of the bridge).
 - **MS-ETS1-2:** Evaluate competing design solutions using a systematic process to determine how well they meet the criteria and constraints of the problem. (Relates to the evaluation aspect of the project).
 - **MS-PS1-4:** Develop a model that describes and predicts changes in particle motion, temperature, and state of a pure substance when thermal energy is added or removed. (While not directly related, the behavior of fluids under pressure has connections to particle motion).
 - **MS-PS2-2:** Plan an investigation to provide evidence that the change in an object's motion depends on the sum of the forces on the object and the mass of the object. (Forces involved in the movement of the bridge).
- o **ISTE Standards for Students:**
 - **Empowered Learner (1):** Students leverage technology to take an active role in choosing, achieving and demonstrating competency in their learning goals, informed by the learning sciences. (Through project creation and problem-solving).
 - **Innovative Designer (4):** Students use a variety of technologies within a design process to identify and solve problems by creating new, useful or imaginative solutions. (The core of the bridge project).
 - **Computational Thinker (5):** Students develop and employ strategies for understanding and solving problems in ways that leverage the power of technological methods to develop and test solutions. (The systematic approach to building and testing the bridge).

Time Frame :90 mins

Objectives: -

- Identify what a breadboard is
- Build a circuit on a breadboard
- Identify what the LDR is

Challenges questions for this lesson: -

- What was the most significant difficulty you encountered while building or integrating the light-sensing circuit into your bridge?
- How did you attempt to solve this challenge? What steps did you take or who did you ask for help?
- What is one key lesson you learned about problem-solving or electronics from overcoming (or not overcoming) this challenge?

SEL relation: -

Self-Awareness &

Self-Management: Reflect on your feelings during successes and challenges, and the strategies you used for planning, staying focused, and troubleshooting.

Collaboration &

Communication: Consider how you interacted with peers and the teacher, sharing ideas, offering/receiving help, and communicating your process.

Responsible Decision-Making:

Think about the safety choices you made and how you considered the functionality and design of your project.

1. Engage

- **Teacher Role:**
 - Facilitates a class discussion.
 - Introduces a real-world problem through a story.
 - Asks open-ended questions to encourage student curiosity.
 - Asks initial questions to gather student ideas.
 - Guides students to connect the story to potential solutions.
- **Expected From Students:**
 - Actively listen to the story.
 - Participate in the class discussion.
 - Share their initial thoughts and ideas about how to solve the problem.
 - Ask clarifying questions.
 - Brainstorm potential materials.
- **Parts of the books included:**



Laila

After we design a movable bridge, I think we have another problem—how do we light it up at night?

Mrs. Sara

It would be great if we could light it only at night to save energy. We will search together to find a way to do that.

Strategies in this part

🎬 Intro Story: Strategy: Storytelling with a Hook (Narrative Introduction)

- **Implementation:**

- Begin by saying: "Imagine Laila and her friend just finished an amazing project – a movable bridge!"
- Continue with: "They were so proud as it went up and down smoothly."
- Introduce Laila's idea: "But then, Laila had an idea. She said to Mrs. Sara, 'Wouldn't it be cool if our bridge had lights, but only turned on when it gets dark, like streetlights?'"
- Describe Mrs. Sara's response: "Mrs. Sara thought it was a fantastic idea! She told Laila and her friend, 'That's a great challenge! Let's work together to figure out how we can make our bridge smart enough to know when it's nighttime.'"
- Explain their next step: "So, Laila, her friend, and Mrs. Sara started their search for clues and information."
- Pose an initial question to the class: "Just like Laila and Mrs. Sara, we are going to become detectives today! Our mission is to figure out how to make a light turn on automatically when it gets dark. What are some things you think might be involved in sensing light or darkness?"
- Pause for initial, quick student responses.

🎬 Story Reflection: Strategy: Think-Pair-Share followed by Whole Class Discussion (Collaborative Reflection)

- **Questions:**

- "What problem did Laila and her friend want to solve?"
- "What did Mrs. Sara suggest they do to find a solution?"
- "What kind of device do you think would be needed to 'sense' when it's dark?"
- **(Transition to the 'Think' part):** "Think individually for a moment: What kind of materials do you think we might need to use to make a light turn *off* when it becomes darker?"

- **Suggested answers (Teacher Guidance):**

- *Problem:* They wanted the bridge lights to turn on automatically at night.
- *Mrs. Sara's Suggestion:* To search together for information. (This sets the stage for the lesson's exploration).
- *Device to sense darkness:* Students might suggest things like sensors, eyes, or even making a guess about a special type of light detector. Acknowledge all reasonable ideas at this stage.

- o *Materials to turn off light when dark (Think Part):* This is designed to be more challenging and might elicit less direct answers. Encourage abstract thinking. Students might suggest things like covers, something blocking the light, a "darkness switch," or something that reacts to the absence of light. This leads into the need for a light-dependent resistor (LDR).
- **Implementation:**
 - o **Think:**
 - Give students 1-2 minutes of quiet individual thinking time.
 - Focus on the reflection questions, especially the "Think" prompt about materials.
 - o **Pair:**
 - Have students pair up with a classmate.
 - Allow 2-3 minutes to discuss their answers and ideas.
 - Encourage them to share their reasoning.
 - o **Share:**
 - Bring the class back together for a whole-class discussion.
 - Ask pairs to share some of their ideas.
 - List student suggestions on the board.
 - Guide the discussion towards the idea of needing a "light sensor" without explicitly giving away the LDR yet.
 - Acknowledge all ideas and gently probe their reasoning.
 - This will naturally lead into the exploration of the LDR in the next part of the lesson.

Here's the "Think" section outlined with bullet points and incorporating strategies for teachers:

2. Think

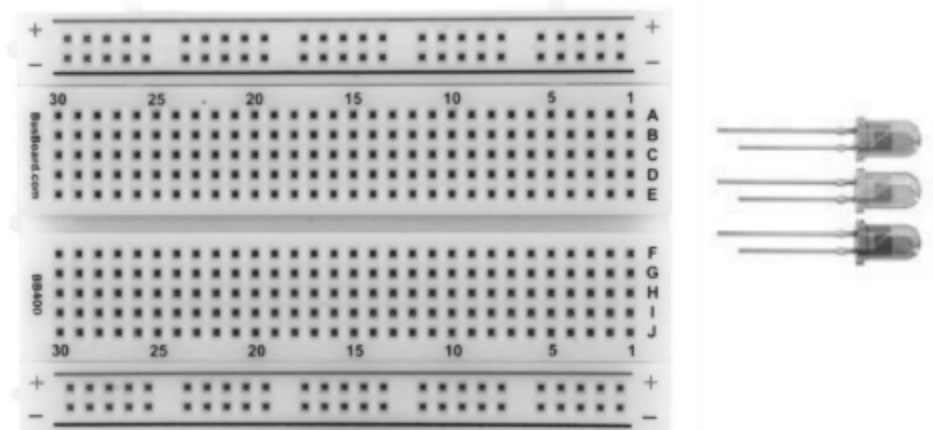
- **Questions and Possible Answers:**
 - o **Question 1:** What do you think are some ways we can detect when it starts to get dark outside?
 - **Possible Answers:**
 - Our eyes can tell.
 - Maybe there's a special kind of sensor.
 - Some animals know when it gets dark.
 - Maybe there are timers.
 - Something that reacts to the amount of light.
 - o **Question 2:** Laila and Mrs. Sara need a component for their bridge that will change its behavior based on the amount of light. What kind of component do you think that could be?
 - **Possible Answers:**
 - A switch that someone has to flip. (Guide them to think about automatic changes).

- Maybe a special kind of light bulb. (Steer towards a light-sensing component).
 - Something that absorbs light.
 - A "light detector."
 - **Question 3:** Think back to our discussion about electricity and circuits. If we want the lights to turn ON when it's dark, how do you think this light-sensing component might work within a circuit?
 - **Possible Answers:**
 - Maybe it closes the circuit when it's dark.
 - Maybe it opens the circuit when it's light.
 - Perhaps it changes how much electricity flows.
- **Brainstorm Web:**
 - **Implementation:**
 - Provide a central circle on a whiteboard or paper labeled "Sensing Darkness."
 - Encourage students to brainstorm related ideas and write them in circles branching out from the center.
 - Draw lines to connect ideas that they think might be related.
 - Prompting questions during the brainstorm:
 - "What things change when it gets dark?" (Less sunlight, shadows, etc.)
 - "How do we know it's dark?" (We can see less, streetlights turn on, etc.)
 - "What kind of things in our everyday lives react to light?" (Solar panels, automatic doors with light sensors, etc.)
 - Allow students to work individually or in small groups for a set time (e.g., 5-7 minutes).
- **Strategies for Teachers:**
 - **Strategy 1: Inquiry-Based Questioning (Facilitating Deeper Thinking)**
 - **Implementation:**
 - Pose open-ended questions that encourage students to think critically and explore possibilities rather than giving direct answers.
 - Use "Why do you think that?" or "Can you explain your reasoning?" to prompt deeper thinking.
 - Avoid immediately correcting wrong answers; instead, encourage other students to share their perspectives.
 - **Strategy 2: Think Time (Promoting Individual Processing)**
 - **Implementation:**
 - After posing a question, explicitly state, "Take a minute to think about this on your own."
 - Allow a silent period of 30-60 seconds before asking for responses. This gives all students a chance to process the question and formulate their ideas.
 - **Strategy 3: (Facilitated Brainstorming)**

- **Implementation:**
 - Use the "Brainstorm Web" activity described above.
 - Actively circulate and encourage participation from all students.
 - Write down all ideas without judgment.
 - Ask clarifying questions to ensure understanding of student contributions.
 - Help students make connections between different ideas on the web.
- **Strategy 4: (Prediction - Observation - Explanation - POE)**
 - **Implementation:**
 - While not a full POE yet, you can start with the prediction aspect. Ask students to predict what kind of component might solve the problem *before* introducing the LDR.
 - Later in the lesson, they will observe the LDR's behavior and then refine their explanations.

2. Explore

- **Teacher Role:**
 - Provide clear and concise instructions for building a simple LED circuit on a breadboard.
 - Circulate around the classroom, offering individual guidance and support as needed.
 - Ensure students are handling components correctly and safely.
 - Encourage students to troubleshoot and problem-solve independently before seeking help.
 - Ask guiding questions to reinforce their understanding of the circuit components and their connections.
 - Facilitate a brief reflection discussion after the activity.
- **Expected From Students:**
 - Follow the step-by-step instructions carefully.
 - Handle the breadboard, LED, and resistor with care.
 - Connect the components correctly according to the instructions.
 - Observe the behavior of the LED once the circuit is complete.
 - Attempt to troubleshoot any issues they encounter.
 - Participate actively in the reflection discussion.
 - Ask questions when they are unsure or encounter difficulties.
- **Parts of the books included:**

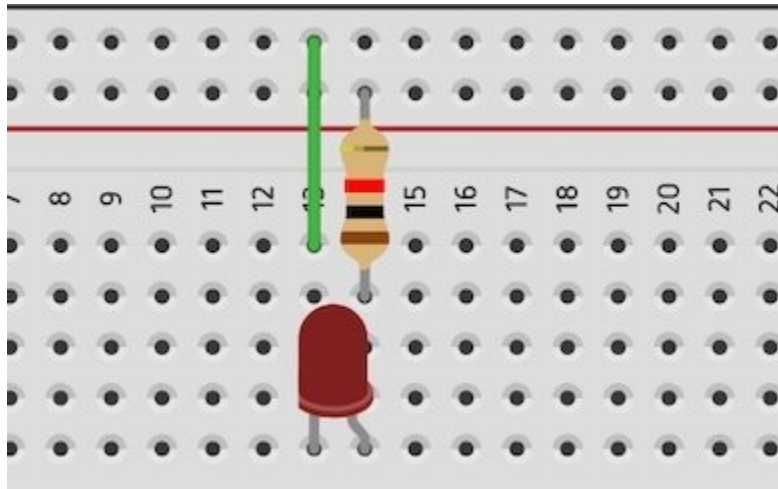


- **Activity - Do It: Building a Simple LED Circuit on a Breadboard**

- **Materials Needed:**

- Breadboard
 - LED (Light Emitting Diode)
 - Resistor (e.g., 220 ohm - check the LED's forward voltage and current requirements)
 - Jumper wires
 - Power source (e.g., 3V battery pack or a regulated power supply set to an appropriate voltage)

- **Step-by-Step Instructions:**



0. **Identify the Components:** Familiarize yourself with the LED (note the longer leg, which is the positive or anode) and the resistor (identify the color bands to determine its value).
1. **Place the LED on the Breadboard:** Insert the two legs of the LED into separate rows on the breadboard, ensuring they are not in the same column. Leave a few rows between them.
2. **Place the Resistor:** Connect one leg of the resistor to the same row as the positive (longer) leg of the LED. Place the other leg of the resistor in a new, empty row.
3. **Connect the Positive Power:** Use a jumper wire to connect the other end of the resistor (the leg in the empty row) to the positive (+) terminal of your power source row on the breadboard (if using a power supply) or directly to the positive terminal of your battery pack.
4. **Connect the Negative Power:** Use another jumper wire to connect the negative (shorter) leg of the LED to the negative (-) terminal of your

power source row on the breadboard or directly to the negative terminal of your battery pack.

5. **Apply Power:** Turn on your power supply or connect the battery pack.
6. **Observe:** The LED should light up. If it doesn't, double-check your connections and the polarity of the LED. Ensure the resistor value is appropriate for your LED and power source.

o **Reflection Questions:**

0. What happened when you connected the power source to your circuit?
1. Why do you think we needed to include a resistor in the circuit with the LED? (Guide them towards understanding current limiting).
2. What do you think would happen if you connected the LED directly to the power source without a resistor? (Discuss the potential for the LED to burn out due to excessive current).
3. Can you identify the path that the electricity flowed in your circuit?
4. What do you think the breadboard helps us do when building circuits?
5. How does this simple circuit relate to the idea of lighting up Laila's bridge? What else would we need to make the lights turn on and off automatically? (These bridges back to the LDR).

3. Explain

• **Teacher Role:**

- o Introduce the Light Dependent Resistor (LDR) as a solution to Laila's problem.
- o Explain the concept of resistance and how it changes with light intensity.
- o Guide students through the provided video and article, highlighting key information.
- o Facilitate discussions to ensure understanding and address misconceptions.
- o Connect the LDR to the broader concept of sensors and their applications.

• **Expected From Students:**

- o Engage actively with the video and article.
- o Share observations and insights from the resources.
- o Ask clarifying questions about the LDR and its function.
- o Relate the LDR back to the problem of the bridge lights.
- o Participate in discussions and contribute to the collective understanding.

- **Parts of the books included:**



- **Watch It: Strategy: Guided Video Analysis (Structured Observation)**
 - **Pre-Watching: Set a purpose.**
 - Pose the question: "As you watch this video, pay close attention to what the LDR looks like and how the presenter explains how it works. Think about how this component could help Laila's bridge lights turn on automatically at night."
 - Briefly introduce the topic of the video (an introduction to LDRs).
 - **During Watching: Active engagement.**
 - Encourage students to take brief notes on key terms, concepts, and observations as they watch.
 - Pause the video at strategic points to ask quick check-in questions, e.g., "What does the video say the LDR is made of?", "How does the resistance change when light shines on it?".
 - **Post-Watching: Discussion and reflection.**
 - **Discussion Prompt 1:** "What did you learn about what an LDR is and how it works from the video?" (Focus on identifying it as a light-sensitive resistor).
 - **Discussion Prompt 2:** "How does the video explain the relationship between light and the LDR's resistance?" (Guide them to understand that more light = lower resistance, less light = higher resistance).
 - **Reflection Question:** "Based on the video, how could an LDR be used in a circuit to turn on the bridge lights when it gets dark?" (Encourage them to think about needing a circuit that activates the lights when the resistance is high).
 - **(Strategy: Think-Pair-Share for Post-Watching Discussion)**
 - **Implementation:** After the video, pose the discussion questions. Have students first think individually about their answers, then discuss with a partner, and finally share their ideas with the whole class. This encourages

participation and allows students to refine their understanding through peer interaction.

- **Read About It: Strategy: Jigsaw Reading (Collaborative Learning)**
 - **Pre-Reading: Discussion.**
 - Briefly introduce the article about LDRs.
 - Ask: "Based on what we saw in the video, what information do you predict this article will provide about LDRs?" (Encourage them to anticipate details about working principles, diagrams, etc.).
 - Review any unfamiliar vocabulary that might appear in the article (e.g., semiconductor, photoconductivity).
 - **During Reading: Engagement.**
 - Divide the article into smaller, manageable sections (e.g., Introduction, Working Principle, Diagram, Applications).
 - Assign each student or small group to read one specific section of the article.
 - Instruct them to focus on understanding the key information in their assigned section and to highlight or note down important points.
 - **Post-Reading: Summarize and reflect.**
 - **Summarize:** Have each student or group explain the main points of their assigned section to the rest of the class. This creates a complete understanding of the article through collaborative sharing. Use visual aids like drawing the LDR symbol or the circuit diagram on the board as students explain.
 - **Reflection Question 1:** "Did the article provide any new information about LDRs that wasn't in the video? If so, what was it?" (Focus on details like the semiconductor material or specific applications).
 - **Reflection Question 2:** "Looking at the circuit diagram in the article, can you explain in your own words how the LDR could be connected to a light bulb and a power source so that the light turns on when it's dark?" (Guide them to understand the basic concept of the LDR controlling the current flow based on light).
 - **(Strategy: Concept Mapping for Post-Reading Reflection)**
 - **Implementation:** After the jigsaw activity, guide the class in creating a concept map on the board or using digital tools. Start with the central concept "LDR (Light Dependent Resistor)." Branch out to include key characteristics, working principles, applications (including the bridge light), and the relationship between light and resistance, drawing information from both the video and the article. This helps students visually organize and connect the information they have learned.

4. Elaborate

- **Teacher Role:** Encourage creativity and application of knowledge.
- **Expected From Students:** Create models, circuits, experiments, or code.
- **Parts of the books included:** Assessment: Focus, Practice, Challenge.

Focus (Answer the questions and list some strategies to apply while answering them.)

Assessment

Focus



What is a breadboard, and how is it organized?

A) A breadboard is a tool used to build electronic circuits. It has a grid of holes where components can be inserted, and the holes are connected internally to allow electricity to flow between them.

B) A breadboard is a type of resistor that changes its resistance based on the amount of light it receives.

C) A breadboard is a tool used to measure the voltage and current in an electronic circuit.

D) A breadboard is a type of LED that turns on when electricity flows through it.

What is the purpose of the resistor in your circuit?

A) The resistor is used to limit the flow of electricity to the LED, preventing it from burning out.

B) The resistor is used to increase the resistance of the LDR when it is exposed to light.

C) The resistor is used to measure the voltage in the circuit.

D) The resistor is used to turn on the LED when the LDR is in darkness.



ity where you built a circuit on
you inserted the components.

r, which is incorrect. Resistors

control current flow, while breadboards facilitate connections.

- o Option C suggests a breadboard measures voltage and current. While you can use tools to measure these *on* a circuit built on a breadboard, the breadboard itself doesn't measure them.
- o Option D suggests a breadboard is an LED. LEDs emit light when current flows through them, which is different from the function of a breadboard.
- **Focus on Keywords:** The question asks "what is a breadboard" and "how is it organized." Option A accurately describes its function as a tool for building circuits and mentions the grid of holes and internal connections, which relates to its organization.

Strategies to Apply While Answering:

- **Recall Prior Knowledge:** Remember the "Explain" and "Explore" activities where the role of the resistor in the LED circuit was discussed. The key reason for the resistor was protection.
- **Eliminate Incorrect Options:**
 - o Option B suggests the resistor increases the resistance of the LDR. The resistor and LDR are separate components with different functions. The resistor's resistance is generally fixed, while the LDR's changes with light.
 - o Option C suggests the resistor measures voltage. A voltmeter measures voltage, not a resistor.
 - o Option D suggests the resistor turns on the LED in darkness. This would be the potential role of an LDR in a more complex circuit, not a standard resistor.

- **Focus on Keywords:** The question asks for the "purpose of the resistor." Option A directly addresses the function of limiting current to protect the LED, which was a key point during the explanation of the simple LED circuit.

Practice: Building a Circuit with LED and LDR

- **Materials Needed:**

- Breadboard
- LED (Light Emitting Diode)
- Resistor (e.g., 220 ohm - for the LED)
- LDR (Light Dependent Resistor)
- Jumper wires
- Power source (e.g., 3V battery pack or a regulated power supply set to an appropriate voltage)

- **Step-by-Step Instructions:**

0. **Place the LED on the Breadboard:** Insert the two legs of the LED into separate rows on the breadboard.
1. **Place the Resistor:** Connect one leg of the resistor to the positive (longer) leg of the LED. Place the other leg of the resistor in a new row.
2. **Place the LDR:** Insert the two legs of the LDR into separate rows on the breadboard.
3. **Connect the Positive Power:** Use a jumper wire to connect the positive (+) terminal of your power source to one leg of the LDR.
4. **Connect LDR to LED/Resistor:** Use another jumper wire to connect the *other* leg of the LDR to the same row as the free end of the resistor (the one not connected to the LED).
5. **Connect the Negative Power:** Connect the negative (shorter) leg of the LED directly to the negative (-) terminal of your power source.
6. **Test in Darkness:** Cover the LDR with your hand or place the circuit in a dark area. Observe the brightness of the LED.
7. **Test in Light:** Expose the LDR to light. Observe the brightness of the LED again.
8. **Record Your Notice:** Note down what you observed about the LED's brightness in both dark and light conditions.

- **Reflection Questions:**

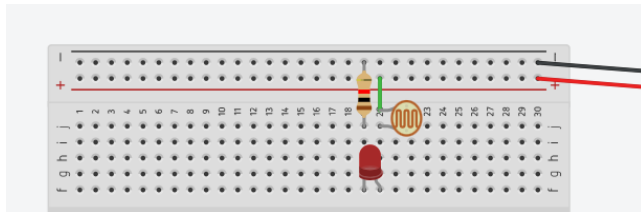
0. What happened to the brightness of the LED when the LDR was in darkness? Why do you think this happened? (Hint: Think about the LDR's resistance in darkness).
1. What happened to the brightness of the LED when the LDR was in light? Why do you think this happened? (Hint: Think about the LDR's resistance in light).
2. How does this circuit demonstrate the LDR's ability to sense light?

3. How could this type of circuit be used in a real-world application, like Laila's bridge lights?

Challenge: How to Get the Reverse Function of LDR?

Challenge: How can you modify the circuit you built in the "Practice" section so that the LED turns **ON** when it is light and turns **OFF** when it is dark? Explain your reasoning with possible circuit modifications.

Possible Answer Strategies (for students):



- **Think About Current Flow:** The current needs to flow through the LED to light it up. The LDR is controlling the resistance in the circuit. How can you arrange the components so that low resistance (in light) allows more current to flow to the LED, and high resistance (in darkness) restricts it?
- **Consider Different Circuit Configurations:** Instead of the LDR being directly in series with the LED and resistor, could it be used in a different way to control the current? Think about how a switch works – it can either complete or break a circuit.
- **Explore the Idea of a Voltage Divider:** An LDR and a fixed resistor can create a voltage divider. The voltage at the junction of these two components changes with the light level. Could this changing voltage be used to control a transistor (which acts as an electronic switch) to turn the LED on or off?
- **Possible Circuit Modifications (Leading to the reverse function):**
 1. **LDR as a Pull-Down Resistor:** Connect the LDR between the positive power supply and a point in the circuit. Connect a fixed resistor between that same point and the negative power supply. Connect the LED (with its own current-limiting resistor) between the positive power supply and the point where the LDR and fixed resistor meet. In this configuration, when light is high (LDR resistance low), the voltage at the connection point will be higher, potentially turning the LED ON (depending on the voltage and other components). When it's dark (LDR resistance high), the voltage at the connection point will be lower, potentially turning the LED OFF.
 2. **Using a Transistor as a Switch:** The changing resistance of the LDR can be used to control the base of a transistor. The transistor can then act as a switch to turn the LED on or off. This requires additional components (a transistor and possibly other resistors to bias the transistor correctly). The LDR could be used in a voltage divider configuration to provide the varying voltage to the transistor's base.

You can use the steps for both circuits from this link

<https://docs.google.com/document/d/1DRgbv0kgXd2dg5axnUAIi961cK0CaLXH/edit?usp=sharing&oid=112295837902879133584&rtpof=true&sd=true>

5. Evaluate

- **Teacher Role:**

- *Facilitate assessments and provide feedback on the integration of the light-sensing circuit into the movable bridge project.

- *Observe students' ability to connect their understanding of the LDR circuit to the functionality of their bridge.

- *Provide guidance on troubleshooting and refining their designs.

- **Expected From Students:**

- *Integrate the LDR and LED circuit into their movable bridge project so that the lights illuminate under low-light conditions.

- *Demonstrate their understanding of how the circuit works in the context of their project. Explain their design choices and any challenges they faced during the integration process.

- **Parts of the books included:** Showcase, Now I can... (as a self-assessment tool).

Showcase: Evaluate the students' work in integrating the light-sensing circuit into the movable bridge and add a rubric for teachers.

To evaluate the students' work in making their bridge light up at night, consider the following rubric:

Movable Bridge with Night Light Integration Rubric

Criteria	Excellent (4 points)	Proficient (3 points)	Developing (2 points)	Needs Improvement (1 point)
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Functionality of Night Light	The lights on the bridge reliably turn ON in low-light conditions (simulating night) and turn OFF or are significantly dimmer in brighter conditions.	The lights on the bridge mostly turn ON in low-light conditions and show some change in brightness in brighter conditions, but may be inconsistent.	The lights on the bridge show a minimal or inconsistent response to changes in light levels.	The lights on the bridge do not respond to changes in light levels.
Integration with Bridge Structure	The light circuit is seamlessly and securely integrated into the bridge structure without hindering its primary function (movement). Wiring is neat and organized.	The light circuit is integrated into the bridge structure, but there may be some minor issues with stability or wiring. The bridge's movement is mostly unaffected.	The light circuit is attached to the bridge, but it is unstable, poorly integrated, or significantly hinders the bridge's movement. Wiring is messy.	The light circuit is not effectively integrated with the bridge structure or prevents the bridge from functioning.
Application of LDR Principles	Clearly demonstrates an understanding of how the LDR functions to control the lights based on light levels within the integrated system. Explanation is accurate.	Shows a basic understanding of how the LDR influences the lights' behavior in the integrated system. Explanation is generally correct.	Demonstrates a limited understanding of how the LDR affects the lights in the integrated system. Explanation may contain inaccuracies.	Shows little to no understanding of how the LDR is intended to control the lights in the integrated system. Explanation is unclear or incorrect.

Circuit Construction & Stability	The LDR and LED circuit on the bridge is well-constructed, with secure connections. Components are protected and positioned thoughtfully.	The circuit on the bridge is functional, but some connections may be slightly loose or components are not optimally positioned.	The circuit on the bridge has several loose connections and appears fragile. Components are haphazardly placed.	The circuit on the bridge is poorly constructed and unreliable. Many connections are loose or incorrect.
Explanation of Design & Challenges	Clearly explains the design choices for integrating the lights and the LDR, and articulates any challenges faced and how they were addressed.	Explains the basic design choices and mentions some challenges encountered during integration.	Provides a limited explanation of the design and struggles to articulate the challenges faced.	Unable to adequately explain the design choices or the challenges encountered.

Scoring:

- 17-20 points: Excellent understanding and execution.
- 13-16 points: Good understanding and execution.
- 9-12 points: Developing understanding and execution.
- 5-8 points: Needs significant improvement.

Relation to Standards:

To link this project to relevant educational standards, consider connections to:

- **Science Standards:**
 - Understanding the properties of light and how sensors can detect changes in light intensity.
 - Basic principles of electricity and circuits (current, voltage, resistance).
 - The function of electronic components like LEDs and resistors.

- **Technology/Engineering Standards:**
 - Applying the engineering design process (design, build, test, evaluate, improve).
 - Understanding and utilizing electronic components for a specific purpose.
 - Troubleshooting and problem-solving in technological design.
 - Safe and effective use of tools and materials.
- **Mathematics Standards:**
 - Potentially analyzing data related to light levels and circuit behavior (if measurements are taken).

Example Connections to Potential Standards (Illustrative - Specific standards vary by region/curriculum):

- **Next Generation Science Standards (NGSS):**
 - **MS-PS4-2:** Develop a model to describe that light reflects from objects and that objects can be seen when light reflected from them enters the eye. (Indirectly related to understanding light).
 - **MS-ETS1-1:** Define the criteria and constraints of a design problem with sufficient precision to ensure a successful solution, taking into account relevant scientific principles and potential impacts on people and the natural environment that may limit possible solutions. (Designing the integrated system).
 - **MS-ETS1-3:** Analyze data from tests to determine similarities and differences among several design solutions to identify the best characteristics of each that can be combined into a new solution to better meet the criteria for success. (Evaluating and improving the integrated bridge and light system).
- **ISTE Standards for Students:**
 - **Innovative Designer (4):** Students use a variety of technologies within a design process to identify and solve problems by creating new, useful or imaginative solutions. (Integrating the light sensor into the bridge).
 - **Computational Thinker (5):** Students develop and employ strategies for understanding and solving problems in ways that leverage the power of technological methods to develop and test solutions. (Troubleshooting the circuit and integration).
 - **Creative Communicator (6):** Students communicate clearly and express themselves creatively for a variety of purposes using the platforms, tools, styles, formats and digital media appropriate to their goals. (Explaining their design and challenges).

UNIT 02 [Intermediate]

↳ Chapter 01 : Programming.

LESSON 01

Make Your Bridge smart.

Time Frame :90 mins

Objectives: -

- ☐ What is an Arduino.
- ☐ Writing simple Arduino sketches (code).
- ☐ Understand Basic knowledge of digital and analog pins.
- ☐ Simulate circuit on Tinkercad.

Challenges questions of this lesson: -

Tell me about a time you had trouble with the traffic light or LED/buzzer project and what you did to solve it. What did you learn about yourself as a problem-solver?

Can you describe another way you could make the LED turn off when the buzzer activates in Tinkercad, using a different circuit idea?

Why is good design important for something like a real traffic light? What could happen if it's designed badly?

SEL Relation: -

Persistence (Self-Management): The "Practice" and "Challenge" activities inherently require students to persevere through difficulties they might encounter while building circuits and potentially writing code.

Problem-Solving Reflection (Self-Awareness): Students will likely face challenges and need to reflect on their approach to identify what works and what doesn't, fostering awareness of their problem-solving strategies.

Responsible Design (Responsible Decision-Making): Students need to make thoughtful choices about how they design their circuits in Tinkercad to achieve the desired functionality safely and logically within the simulation environment.

1. Engage

- **Teacher Role:**

- Initiates a discussion by drawing a parallel between the human brain and the "brain" of a computer.
- Presents a relatable scenario involving a student (Laila) connecting a new concept to a current project.
- Poses a thought-provoking question to the students to bridge their understanding of a CPU to their own project.
- Introduces Arduino as a potential "brain" for their project, sparking curiosity.

- **Expected From Students:**

- Listen attentively to Mrs. Sara's explanation about the CPU.
- Consider the analogy of the CPU as the "brain" of a computer.
- Reflect on their existing bridge project and how a "brain" might be involved in its movement.
- Show curiosity and interest in the introduction of Arduino.
- Begin to think about how a programmable component like Arduino could control movement in their project.

- **Parts of the books included:**



- **Intro Story: Strategy: Analogy and Questioning (Relating New Concepts to Prior Knowledge)**
 - **Implementation:**
 - Begin by saying, "Think about how your own brain helps you decide to move your arm or leg. It sends signals and tells your muscles what to do. In the world of computers, there's something similar called the CPU. Mrs. Sara in our story describes it as a 'super smart helper inside the computer...like the brain that tells everything in the computer what to do!'" (Show the image of Mrs. Sara explaining to the students).
 - Pause and ask the class: "Can you think of any ways a 'brain' could be useful for a project that needs to move, like the bridge Laila and her team are building?" (Allow for a few initial, quick responses).
- **Story Reflection: Strategy: Think-Pair-Share (Connecting to the Project)**
 - **Questions:**
 - "In the story, Laila asks a very important question about their bridge project. What was her question?"
 - "Why do you think Laila connected the idea of the CPU to their bridge's movement?" (Suggested answer: Because movement requires control and decision-making, similar to what a brain does).

- "Mrs. Sara then introduces something new called Arduino. Based on what she says ('It will be the brain of our project and help control how our bridge moves!'), what do you think Arduino might be able to do?" (Suggested answer: Control motors, activate mechanisms, make decisions about when and how the bridge moves).
- **Implementation:**
 - **Think:** Give students 1-2 minutes to individually consider the reflection questions.
 - **Pair:** Have students pair up to discuss their thoughts and share their initial ideas about what Arduino might be.
 - **Share:** Bring the class back together and ask pairs to share their ideas about what Arduino could do to help control the bridge's movement. List their ideas on the board.

Think Question: How could the Arduino help us to solve the problem with our project (making the bridge move)?

Possible Answers (Teacher Guidance):

- Arduino is like a mini-computer that we can program.
- We can write instructions (code) for the Arduino to tell motors or other parts of our bridge when and how to move.
- Arduino can receive information from sensors (like buttons or maybe even the light sensor we used before) and make the bridge move based on that information.
- It could control the speed or direction of the movement.
- It could automate the movement of the bridge, so we don't have to manually push or pull it.

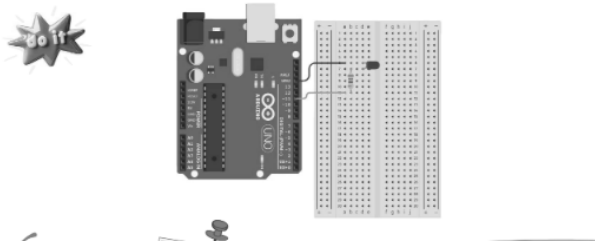
2. Explore

- Teacher Role:

- Introduce the task of making an LED blink using an Arduino, referring to the provided circuit diagram.
- Explain that they will need to write simple code (an "Arduino sketch") to control the digital output pin connected to the LED.
- Guide students on how to identify the components in the diagram: the Arduino Uno, the breadboard, the LED, and the resistor.
- Explain the connection between the Arduino's digital output pin (indicated in the diagram) and the LED on the breadboard, emphasizing the role of the resistor.
- Facilitate a discussion on the basic logic of blinking: turning the LED ON for a short time, then OFF for a short time, and repeating this process.
- Provide initial guidance on the structure of a basic Arduino sketch, mentioning the `setup()` and `loop()` functions.
- Encourage students to think about the commands needed to control a digital pin (e.g., `digitalWrite()`) and to introduce a delay (`delay()`).
- Offer support and answer questions as students begin to write their code.
- Expected From Students:
 - Examine the provided circuit diagram carefully to understand how the LED and resistor are connected to the Arduino.
 - Identify the Arduino Uno board, the breadboard, the LED, and the resistor in the diagram.
 - Understand that they need to write code for the Arduino to control the LED.
 - Begin to brainstorm the sequence of commands needed to make the LED blink (ON, wait, OFF, wait, repeat).
 - Start writing a simple Arduino sketch using the `digitalWrite()` and `delay()` functions (or equivalent if using a simulation environment).
 - Ask clarifying questions about the Arduino syntax or the circuit connections.
- Parts of the books included:

Explore

Here's a circuit with an LED and a resistor connected to an Arduino. Write the code to make the LED blink.



- Activity - Do It: Make an LED Blink with Arduino
 - Materials Needed (Based on the Image):
 - Arduino Uno R3 (or compatible board)
 - Breadboard
 - LED
 - Resistor (appropriate value for the LED, likely in the range of 220Ω to $1k\Omega$)
 - Jumper wires (as indicated in the diagram to connect the Arduino to the breadboard)
 - Computer with Arduino IDE (or access to a Tinkercad simulation environment as mentioned in previous objectives)
 - USB cable for connecting the Arduino to the computer (if using a physical board)

Step-by-Step Instructions (Conceptual - Code needs to be written):

- Identify the Connected Pin: Observe the circuit diagram to see which digital pin on the Arduino is connected to the LED (via the resistor). Let's assume for this example it's Digital Pin 13 (often used for built-in LEDs or simple external circuits).
- Open the Arduino IDE (or Tinkercad): Launch the Arduino programming environment on your computer or open the Tinkercad Circuits simulator.

- Create a New Sketch: Start a new Arduino sketch (code file).
- Set the Pin Mode in `setup()`: In the `setup()` function, you need to tell the Arduino that the digital pin connected to the LED will be used as an output. Use the `pinMode()` function. For example: `pinMode(13, OUTPUT);`
- Write the Blinking Logic in `loop()`: In the `loop()` function, which runs repeatedly, write the code to turn the LED ON, wait for a short time, turn the LED OFF, and wait again. Use the `digitalWrite()` function to control the pin's state (HIGH for ON, LOW for OFF) and the `delay()` function to pause for a specified number of milliseconds. For example:
Arduino

```
digitalWrite(13, HIGH); // Turn the LED ON
```

```
delay(1000);           // Wait for 1000 milliseconds (1 second)
```

```
digitalWrite(13, LOW); // Turn the LED OFF
```

```
delay(1000);           // Wait for 1000 milliseconds (1 second)
```

- Connect the Arduino (if using physical hardware): Connect your Arduino board to your computer using the USB cable.
- Select the Board and Port: In the Arduino IDE, go to "Tools" > "Board" and select "Arduino Uno" (or your specific board). Then, go to "Tools" > "Port" and select the serial port that your Arduino is connected to.
- Upload the Code: Click the "Upload" button in the Arduino IDE to send the code to your Arduino board. If using Tinkercad, start the simulation.
- Observe the LED: The LED connected to the Arduino should now be blinking.

Reflection Questions:

- What happens to the LED when the `digitalWrite()` function sends `HIGH` to the pin? What about when it sends `LOW`?
- What does the `delay()` function do in your code? How does changing the value inside `delay()` affect the blinking speed?
- What is the difference between the `setup()` function and the `loop()` function in an Arduino sketch? Why is the blinking code placed in `loop()`?

- How does the code you wrote control the flow of electricity to the LED?
- How could you modify this code to make the LED blink faster or slower?
- How does this simple example demonstrate the Arduino's ability to control external components?

1. Explain

Teacher Role:

- Introduce **basic Arduino concepts** (what is Arduino, what it does).
- Explain the **role of sensors and outputs** (e.g., LDR as input, LED as output).
- Guide students through the **Engineering Design Process** (Ask, Imagine, Plan, Create, Test, Improve).
- Use questioning techniques like *Think-Pair-Share* to encourage participation.
- Facilitate the integration of *video* and *text* as learning resources.

Expected From Students:

- Actively listen, ask questions, and share what they know or observe.
- Participate in discussions about sensors and real-life examples.
- Connect new knowledge to previous STEAM experiences.
- Share reflections after watching and reading.
- Engage in hands-on experimentation with the LDR-LED circuit.

Parts of the Books Included:

If you are using a STEAM textbook or Arduino guide for kids, this would relate to:

- Chapter on *Inputs and Outputs*

- Activity: *Building a Light-Activated LED*
 - Vocabulary terms: *Sensor, LED, Code, Upload, Input, Output*
-

Watch It: <https://www.youtube.com/watch?v=CSx6k-zXILE>

Pre-Watching: Set a Purpose

- Strategy: **K-W-L Chart**

Ask students:

- What do you *Know* about sensors?
- What do you *Want* to know?

Complete the *Learned* section after the video.

✓ *Implementation:* Draw the chart on the board or use handouts.

During Watching: Active Engagement

- Strategy: **Guided Video Notes**

Students fill in blanks or answer specific questions while watching.

Example prompts:

- What components are used?
- What triggers the LED to turn on?

✓ *Implementation:* Provide a worksheet or use Google Forms.

Post-Watching: Discussion and Reflection

- Strategy: **Turn and Talk + Whole-Class Share**

Students turn to a partner and discuss what surprised them.

✓ *Implementation:*

- Ask: "Why do you think the LED turns on in the dark?"
 - Highlight the concept of **resistance** in the LDR.
-

Read About It: <https://www.create-learn.us/blog/arduino-programming-for-kids/>

Pre-Reading: Discussion

- Strategy: **Connect to Prior Knowledge**

Ask:

- “Have you ever heard of programming before?”
 - “What devices in your home might use something like Arduino?”
- ✓ *Implementation:* Class discussion or mini brainstorming session.

During Reading: Engagement

- Strategy: **Jigsaw Reading**

Divide article into sections:

1. What is Arduino?
2. Why is it great for kids?
3. What can you build with it?

Each group presents their part.

✓ *Implementation:* Small groups with shared summaries.

Post-Reading: Summarize and Reflect

- Strategy: **One-Sentence Summary + Exit Ticket**

Ask each student to write one sentence summarizing what they learned.

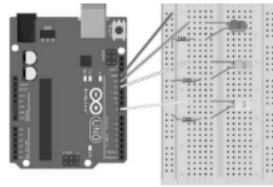
✓ *Implementation:* Collect on sticky notes or a shared Padlet wall.

4. Elaborate

- **Teacher Role:** Encourage creativity and application of knowledge.
- **Expected From Students:** Create models, circuits, experiments, or code. The "Challenge" section explicitly asks students to "Create a circuit where a LED turns off simultaneously with the activation of a buzzer on tinkercad." This requires them to apply their understanding of basic circuit concepts and Tinkercad simulation.
- Parts of the books included:

Practice

Try to make a traffic light on Arduino using 3 LEDs and 3 resistors and write the code on Arduino IDE.



- Assessment:
 - Focus:

Assessment

Focus

What is Arduino primarily used for?

- A) Cooking food.
- B) Controlling electronics.
- C) Playing music.
- D) Writing stories.

What are the three main components of Arduino?

- A) Software, hardware, and battery.
- B) Microcontroller, IDE, and code.
- C) Inputs, outputs, and sensors.
- D) Circuits, wires, and lights.

Strategies to apply while answering them.

- Carefully read each question to understand what is being asked.
- Review prior learning about Arduino's core functionality and components.
- Eliminate incorrect answer choices based on your understanding.

- For the first question, consider the broad applications of Arduino beyond the other options.
- For the second question, recall the fundamental elements required to program and use an Arduino board.
- Practice:

- **Materials Needed:**

- Arduino board
- 3 LEDs (likely red, yellow, green)
- 3 resistors (appropriate value for the LEDs)
- Breadboard
- jumper wires
- computer with Arduino IDE.

- **Step-by-Step Instructions:**

- Left for the student to devise and implement in the Arduino IDE and on the breadboard.
- This is the core of the "Practice" activity, requiring students to plan and execute the circuit and code.

- **Reflection Questions:**

- What challenges did you encounter while building the traffic light circuit?
- How did you approach writing the code to control the LEDs?
- What role do the resistors play in the circuit?
- How could you modify the code to change the timing of the traffic light?

- **Challenge:**

Left for the students to be made at home. This implies that the design and implementation of the LED-buzzer circuit are entirely student-driven.

2. Evaluate

- **Teacher Role:**

- Facilitate assessments and provide feedback.
- This involves observing students during the "Practice" activity, reviewing their Arduino code and circuit implementations for the traffic light, and assessing their final "Challenge" project.

- **Expected From Students:**

- Answer questions (in the "Focus" section), demonstrate understanding (through the "Practice" activity), and make projects (the "Challenge").

- **Parts of the books included:**

- **Focus:** Assesses basic recall and understanding of Arduino fundamentals.
- **Practice:** Evaluates the student's ability to apply their knowledge to a practical task, including circuit building and coding.
- **Challenge:** Provides an opportunity for students to demonstrate a deeper understanding and problem-solving skills by designing and implementing their own circuit based on the given criteria.

Showcase: Evaluate the students' work in the main project (Challenge) and add a rubric so teachers can follow.

Here's a rubric that teachers can use to evaluate the students' work on the "Challenge" project (LED turns off with buzzer activation):

Criteria	Excellent (4 points)	Proficient (3 points)	Developing (2 points)	Needs Improvement (1 point)
Circuit Design (Tinkercad)	Circuit is logically sound and	Circuit is mostly correct with minor	Significant errors in circuit design	Circuit design is largely incorrect or

	correctly implemented in Tinkercad. LED and buzzer are appropriately connected.	errors in connections. LED and buzzer are generally connected.	and connections. LED and/or buzzer connection is flawed.	incomplete. LED and buzzer are not correctly connected.
Functionality (Simulation)	LED turns OFF precisely when the buzzer is activated in the Tinkercad simulation.	LED turns off mostly simultaneously with the buzzer, with a slight delay.	LED and buzzer activation are not clearly synchronized, or one component doesn't function as expected.	LED does not turn off, or the buzzer does not activate upon the intended trigger.
Understanding of Components	Demonstrates a clear understanding of how the LED and buzzer work within the circuit. Explanation is accurate.	Shows a general understanding of the LED and buzzer's function in the circuit. Explanation is mostly correct.	Demonstrates a limited understanding of the LED and buzzer's role. Explanation contains some inaccuracies.	Shows little to no understanding of how the LED and buzzer function in the circuit. Explanation is unclear or incorrect.
Creativity & Problem Solving	Shows a creative approach or efficient solution to	Applies standard methods to achieve the outcome.	Follows a basic approach with limited creativity.	Approach is unclear or ineffective. Unable to

	achieve the desired outcome. Demonstrate s effective problem-solving if issues arose.	Shows some effort in troubleshooting minor issues.	Struggles to resolve issues independently.	resolve basic issues.
Documentation (if required)	Clear and concise explanation of the circuit design and how it achieves the desired functionality. (If documentation was part of the assignment).	Adequate explanation of the circuit design and functionality. (If documentation was part of the assignment).	Explanation is unclear, incomplete, or contains errors. (If documentation was part of the assignment).	Little to no explanation provided or is entirely inaccurate. (If documentation was part of the assignment).

Relation to Standards:

The activities in these images relate to several educational standards, particularly in areas like:

- **Science and Technology:** Understanding basic electrical circuits, components like LEDs and buzzers, and the principles of input and output in electronic systems.
- **Engineering Design Process:** Encouraging students to plan, create, test, and refine their circuit designs in Tinkercad and potentially in a physical implementation. The "Challenge" specifically aligns with the "Create" stage of the EDP.

- **Computer Science/Computational Thinking:** If students are expected to write code to control the buzzer and LED (though not explicitly stated in the "Challenge"), this would relate to basic programming concepts and logic.
- **ISTE Standards for Students:** Particularly related to:
 - **Empowered Learner:** Students use technology to seek feedback and demonstrate their learning in creative ways.
 - **Digital Citizen:** Students understand and practice safe and responsible use of technology (in this case, simulation software).
 - **Knowledge Constructor:** Students critically curate resources, create artifacts, and make meaningful learning experiences for themselves and others.
 - **Innovative Designer:** Students use a variety of technologies within a design process¹ to create innovative solutions.
 - **Computational Thinker:** Students develop and employ strategies for understanding and solving problems in ways that leverage the power of² technological methods.

To find all the steps on details visit this document

<https://docs.google.com/document/d/1YTrs39ixA1rrMXHIBjXLEbBGQBcFFV5j/edit?tab=t.0>

Time Frame :90 mins

Objectives: -

- Design and build separate circuits to control servo motor.
- Understand the sensors especially water level sensor and how to function in our project.

Challenges questions: -

- How can you adjust your code or design if the bridge needs to lift higher?
- How can you make the bridge lift automatically during a flood?
- What real-life materials would you choose for a permanent bridge, and why?

SEL Relation: -

- **Responsible Decision-Making:**
Students identify real-world problems (flooding) and propose safe engineering solutions (lifting bridges).
- **Teamwork & Communication:**
Students collaborate on coding, building, and testing their bridge, learning to listen and contribute effectively.
- **Growth Mindset & Resilience:**
Students learn through trial and error, improving their designs and code with persistence.

1. Engage

- **Teacher Role:**

- Facilitate discussion by asking thought-provoking questions.
 - Encourage critical thinking and idea sharing.
 - Guide students through reflecting on real-world problems.
 - Use visuals and storytelling to stimulate curiosity.
 - Connect the story to students' prior knowledge and experiences.
-

• **Expected From Students:**

- Actively listen to the story.
 - Share opinions and possible solutions.
 - Reflect on real-world issues and how they relate to engineering challenges.
 - Answer questions and collaborate in discussion.
 - Begin thinking about how they can solve the problem as engineers.
-

• **Parts of the Book Included:**



• **Intro Story Strategy:**

Strategy Name: *Storytelling with Visual Prompts*

How to Implement:

- Show the comic strip or printed visuals to the class.
 - Read the dialogue aloud, using character voices for engagement.
 - Pause and ask students how the characters might feel.
 - Link the story to real-life events (e.g., recent floods or bridge failures in the news).
 - Ask: “What would *you* do if you were the engineer in this situation?”
-

• Story Reflection:

Questions:

1. Why did the bridge collapse in Adam’s story?
2. What kind of problem could flooding cause for our bridge?
3. What was Laila’s suggestion?
4. Do you think it’s a good idea? Why?
5. How could we make a bridge that can lift and lower?

Suggested Answers:

1. Because of flooding.
 2. It could damage or collapse the bridge.
 3. To lift the bridge during a flood.
 4. Yes, because it helps protect the bridge and lets water pass safely.
 5. By using motors and a control system that can lift the bridge.
-

• Reflection Strategy:

Strategy Name: *Think-Pair-Share*

How to Implement:

- Ask students to think individually about how they would solve the problem.
- Pair them with a classmate to share their ideas.
- Let pairs share with the class to gather a range of solutions.
- Summarize and connect their ideas to engineering principles.

Think

• Questions and Possible Answers:

1. **What are the electrical components that we need to add to the bridge to make the bridge lift and lower?**

Possible Answers:

- Servo motor (to lift/lower the bridge)
- Arduino board (to control the motor)
- Jumper wires (to make connections)
- Power source (USB or battery)
- Sensors (optional – to detect water level or approaching boats)
- Button or switch (to manually activate the bridge if needed)

2. **Do you think we will use an Arduino board in this step? Why?**

Possible Answer:

- Yes, because the Arduino can control the servo motor by receiving code instructions. It acts like the brain of the system and helps us program when and how the bridge should move.

• Brainstorm Web:

You can draw this on the board or give students a printed copy to fill out:

Center Idea: *“How can we lift the bridge using electronics?”*

Branches:

- **Movement Mechanism:** Servo motor
 - **Controller:** Arduino Uno
 - **Power:** USB or Battery
 - **Control Signal:** Code/Program
 - **Input:** Button or Sensor
 - **Safety Feature:** Stop at certain angle, emergency switch
 - **Real-Life Use:** During floods, for boats, when road is closed
-

• **Strategy for Delivery:**

Strategy Name: *Brainstorm Carousel*

How to Implement:

1. Divide the class into small groups.
2. Give each group a large sheet with a central idea (e.g., “Bridge that lifts”).
3. Ask each group to write or draw their ideas around the topic.
4. After 3–5 minutes, rotate the sheets to the next group.
5. Each group adds new ideas or builds on previous ones.
6. Rotate until each group has seen all the brainstorm.
7. Share final webs with the class and discuss patterns.

Alternative Strategy: *Think Aloud & Visual Mapping*

- Use a whiteboard or smartboard to model your thinking out loud.
 - Ask students to suggest ideas as you build the web together.
 - This encourages idea-sharing and logical connection-building.
-

2. Explore

• Teacher Role:

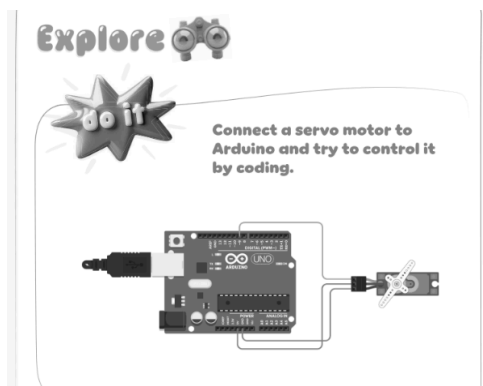
- Guide students through setting up the servo motor with the Arduino.
 - Ensure students understand the function of each component.
 - Support students in writing and uploading basic Arduino code.
 - Facilitate troubleshooting and encourage collaborative problem-solving.
 - Monitor safety and correct usage of electrical components.
-

• Expected From Students:

- Follow instructions to connect the servo motor to the Arduino board.
- Participate in coding and testing the servo motor's motion.
- Observe how the servo reacts to code and make adjustments.
- Collaborate in teams and discuss observations.
- Reflect on how this activity relates to lifting a bridge in the real world.

• Parts of the Book Included:

“Connect a servo motor to Arduino and try to control it by coding.”



• Activity – Do It:

Materials Needed:

- 1 Arduino Uno board
 - 1 Servo motor
 - Jumper wires
 - Breadboard (optional)
 - USB cable for Arduino
 - Laptop with Arduino IDE installed
-

Step-by-Step Instructions:

1. Connect the Servo Motor:

- Connect the **orange/yellow wire (signal)** of the servo to **digital pin 9** on the Arduino.
- Connect the **red wire (VCC)** to the **5V** pin on the Arduino.
- Connect the **brown/black wire (GND)** to the **GND** pin.

2. Write the Code:

- Open the Arduino IDE.
- Use the following sample code:

```
#include <Servo.h>
```

```
Servo myServo;
```

```
void setup() {  
  myServo.attach(9);  
}
```

```
void loop() {  
  myServo.write(0); // move to 0 degrees
```

```
delay(1000);  
myServo.write(90); // move to 90 degrees  
delay(1000);  
myServo.write(180); // move to 180 degrees  
delay(1000);  
}
```

3. Upload and Test:

- Connect the Arduino to the laptop via USB.
- Upload the code to the board.
- Watch how the servo moves between positions.

Reflection Questions:

1. What happened when you uploaded the code?
2. How does the servo motor move?
3. How could this movement help us build a bridge that lifts?
4. Can you think of other projects where we might use a servo motor?

3.Explain

• Teacher Role:

- Clearly explain the difference between servo motors and DC motors.
- Facilitate group discussions to deepen understanding.
- Guide students in connecting video/article information to their project (the lifting bridge).
- Encourage students to ask questions and share reflections.
- Use visual aids or a real motor to enhance explanations.

• Expected From Students:

- Watch the video actively and take notes on key differences and uses of servo motors.
 - Read the article and highlight important comparisons.
 - Participate in discussions, ask questions, and share new facts.
 - Reflect on how servo motors are important in their bridge project.
-

• **Parts of the Books Included:**



• **Watch It**

Pre-Watching: Set a Purpose

- Teacher says: "We're about to watch a video explaining how servo motors work and why they are useful. Pay close attention to how they move and what makes them different from regular motors."

During Watching: Active Engagement

- Students use a simple note-taking chart with two columns:
 - "What I saw" (observations about the servo)

- “Why it matters” (importance to the bridge project)
 - Teacher pauses the video at key moments to ask guiding questions like:
 - “What do you notice about how the servo turns?”
 - “What kind of signal does the servo need?”
-

Post-Watching: Discussion and Reflection

- Lead a whole-class discussion:
 - “Why do you think a servo is more useful than a DC motor in our project?”
 - “What would happen if we used the wrong type of motor?”
 - Students reflect in journals or exit slips.
-

Suggested Strategy: “See, Think, Wonder”

- **See:** What did you observe in the video?
 - **Think:** What do you think is happening?
 - **Wonder:** What are you curious about now?
-

• Read About It

Pre-Reading: Discussion

- Teacher introduces the topic:
“We’ll read an article comparing servo motors and DC motors. What do you already know about motors? What questions do you have?”
-

During Reading: Engagement

- Students use highlighters or color-coded pencils to:
 - Highlight servo motor features in one color.

- Highlight DC motor features in another.
 - Partner reading can be used to support younger students or ELLs.
-

Post-Reading: Summarize and Reflect

- Students complete a Venn diagram comparing servo and DC motors.
 - Teacher asks:
 - “Which motor is more suitable for our bridge project? Why?”
 - “Can we think of another project where a DC motor might be better?”
-

Suggested Strategy: “Think-Pair-Share”

1. **Think:** Students reflect individually on the main ideas.
2. **Pair:** Share ideas with a partner.
3. **Share:** Discuss ideas as a class to build a collective understanding.

4.Elaborate

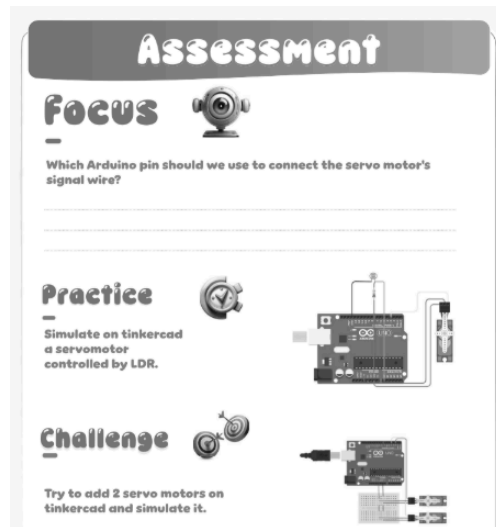
Teacher Role:

- The teacher's role in this "Elaborate" phase is to **encourage creativity and the application of knowledge**.
- They should act as facilitators, guiding students to think beyond the basic concepts and explore more intricate uses of servo motors and Arduino.
- This involves prompting students to think critically, troubleshoot independently, and potentially integrate other components or functionalities into their projects.

Expected From Students:

- Students are expected to **create models, circuits, experiments, or code** that build upon the foundational understanding of servo motors and Arduino.
- The "Challenge" section explicitly asks them to add and simulate two servo motors, indicating an expectation for them to design and implement a more complex circuit and corresponding code in Tinkercad.
- This demonstrates their ability to apply learned concepts in a new, unguided situation.

Parts of the books included:



Fundamental concepts of servo motors:

Their operation, control signals, and wiring.

- **Introduction to Arduino:** Pin types (digital, analog, PWM), basic programming concepts, and the Arduino IDE.
- **Light Dependent Resistor (LDR) basics:** How they work and how to interface them with Arduino.
- **Tinkercad simulation environment:** Basic usage for creating circuits and running simulations.

Assessment: The overall assessment seems to be structured in three stages:

- **Focus:** A direct question to test basic understanding.
- **Practice:** A guided simulation to reinforce a specific application.
- **Challenge:** An independent project to evaluate the ability to apply and extend the learned concepts.

Let's now elaborate on each part of the assessment based on the image:

Focus:

- **Question:** Which Arduino pin should we use to connect the servo motor's signal wire?

- **Answer:** The servo motor's signal wire should be connected to a **digital pin capable of Pulse Width Modulation (PWM)** on the Arduino. These pins are typically marked with a tilde (~). PWM signals are necessary to control the angle of the servo motor by varying the duty cycle of the pulse sent to the signal wire.
- **Strategies to apply while answering:**
 1. **Recall the principle of servo motor control:** Remember that servo motor position is controlled by the duration of a pulse sent to its signal wire.
 2. **Identify Arduino pin types:** Differentiate between digital input/output pins and those that can generate PWM signals.
 3. **Consult Arduino documentation (if needed):** If unsure about which pins support PWM, refer to the Arduino board's pinout diagram or official documentation.
 4. **Consider the function of the signal wire:** Understand that this wire carries the control signal, which in the case of a servo, requires PWM.

Practice:

- **Goal:** Simulate on Tinkercad a servomotor controlled by an LDR.
- **Materials Needed (based on typical LDR-servo control):**
 - Arduino board (Uno or similar)
 - Servo motor
 - Light Dependent Resistor (LDR)
 - Resistor (typically 10k Ω for a voltage divider with the LDR)
 - Breadboard
 - Jumper wires
 - Tinkercad account and access to the Circuits environment.
- **Step-by-Step Instructions (conceptual - specific steps would be performed in Tinkercad):**
 0. **Build the circuit in Tinkercad:**

- Place the Arduino, servo motor, LDR, and resistor on the virtual breadboard.
- Connect the servo motor's power (VCC) and ground (GND) pins to the Arduino's 5V and GND respectively.
- Connect the servo motor's signal wire to one of the Arduino's PWM digital pins (e.g., pin 9).
- Create a voltage divider circuit with the LDR and the resistor. Connect one end of the LDR to 5V, the other end to one end of the resistor. Connect the other end of the resistor to GND.
- Connect the junction point of the LDR and the resistor to an analog input pin on the Arduino (e.g., pin A0).

1. Write the Arduino code in Tinkercad's code editor:

```
#include <Servo.h>
```

```
Servo myServo;
```

```
void setup() {  
  myServo.attach(9); // Attach the servo to pin 9  
}
```

```
void loop() {  
  myServo.write(90); // Rotate to 90 degrees  
  delay(1000);  
  myServo.write(0); // Rotate back to 0 degrees  
  delay(1000);  
}
```

2. **Simulate the circuit:** Click the "Start Simulation" button in Tinkercad.
 3. **Observe the behavior:** Change the virtual light intensity on the LDR in Tinkercad and observe how the servo motor's angle changes in response.
- **Reflection Questions:**
 0. How does the light intensity affect the resistance of the LDR?
 1. How does the change in LDR resistance affect the voltage reading at the analog input pin?
 2. How does the Arduino code map the analog reading to the servo motor's angle?
 3. What happens if you change the resistor value in the voltage divider?
 4. Can you think of real-world applications for a light-controlled servo motor?

Challenge:

- **Task:** Try to add 2 servo motors on Tinkercad and simulate it.
- **Elaboration:** This section is intentionally left for the students to design and implement. It requires them to extend their understanding from controlling one servo to controlling two independently or in a coordinated manner. They will need to:
 - Connect the second servo motor to the Arduino, ensuring it has power, ground, and its signal wire connected to another PWM digital pin.
 - Modify the Arduino code to control both servo motors. This might involve controlling them based on different inputs, making them move in sequence, or coordinating their movements.
 - Simulate the circuit in Tinkercad and observe the behavior of both servo motors.

5.Evaluate

- **Teacher Role:**

- Facilitate the evaluation process by observing students during their project demonstration.
 - Ask guiding questions to assess students' understanding of electronics, programming, and the purpose of the lifting bridge.
 - Provide individualized feedback to encourage growth.
 - Use a rubric to ensure fair and clear assessment.
- **Expected From Students:**
 - Demonstrate their bridge model lifting up and down using the servo motor.
 - Explain how the servo motor is connected to the Arduino.
 - Discuss when and why a bridge might need to be lifted (e.g., during a flood).
 - Reflect on how they programmed and controlled the servo motor.
-

Showcase Activity: Lifting Bridge Demo

Each student or group will present their working model:

- The servo motor must lift and lower a paper/plastic bridge.
 - The Arduino code must function properly (lifting when needed, lowering otherwise).
 - Students should describe how the solution helps during a flood.
 - Students will reflect on what they learned and any challenges they faced.
-

Evaluation Rubric:

Criteria	Excellent (3)	Good (2)	Needs Improvement (1)
Understanding of Servo Motor	Can explain how it works and controls it fully	Can explain with some support	Limited understanding
Arduino Connection	All components correctly connected and explained	Minor mistakes in setup	Major setup issues
Code Functionality	Code runs smoothly and meets the objective	Minor bugs present but working	Code doesn't perform as expected
Bridge Lifting Mechanism	Fully functional and synchronized	Partially functional	Mechanism doesn't work
Teamwork & Communication	Clear roles and excellent explanation	Average participation	Lacks clarity or participation
Creativity & Design	Innovative use of materials	Standard design	Minimal design effort

Relation to Standards:

This project aligns with the following educational standards:

- **NGSS (Next Generation Science Standards):**
 - **ETS1.A:** Defining Engineering Problems
 - **ETS1.B:** Developing Possible Solutions
 - **PS3.C:** Relationship Between Energy and Forces
- **ISTE Standards for Students:**
 - **1.4 Innovative Designer:** Students use a variety of technologies within a design process to identify and solve problems.
 - **1.5 Computational Thinker:** Students develop and employ strategies for

Save Your Smart Bridge

LESSON 02

Time Frame: 90 mins

Objectives:

- Students learn about water level sensors and servo motors, understanding their individual functions and how they can be connected to an Arduino microcontroller.
- Students develop coding skills to read input from a water level sensor and use that data to programmatically control the movement of a servo motor.
- Students apply their knowledge to design and build a project simulating an automatic bridge, where changes in water level trigger a corresponding movement of the servo motor, demonstrating sensor-actuator interaction.

Challenge questions for this lesson:

- How could you potentially integrate this real-world data into your Arduino project to trigger the bridge lift based on actual flood conditions in a specific location?
- How could you create a simple user interface (e.g., using a small LCD screen or a series of LEDs) to display the current water level and the status of the bridge?

teacher role.

SEL Relation:

Self-Awareness: Students can reflect on their strengths and challenges during the project, recognizing their problem-solving skills and areas where they needed to learn more.

Self-Management: Students need to manage their time effectively, organize their materials, and persevere through challenges they encounter while building and coding their project.

Social Awareness: Students can learn about the real-world implications of flood detection and how technology can be used to address safety and environmental concerns, fostering empathy and understanding of community needs.

- The teacher's responsibilities in the "Engage" phase include **facilitating discussions** sparked by the intro story and **asking questions** that encourage students to think about the problem presented and brainstorm potential solutions.
- They should create a welcoming environment where students feel comfortable sharing their initial ideas and observations.

Expected From Students:

- Students are expected to be **participating in discussions** actively, **sharing their initial ideas** about how a flood can be detected, and **asking clarifying questions**.
- They should be attentive to the intro story and reflect on the scenario presented.

Parts of the books included:



Intro Story:

- **Strategy: Problem-Based Introduction (Scenario Presentation)**
- **Implementation:** The teacher can begin by presenting the scenario depicted in the image through storytelling or by displaying the image itself. The core of the story is Adam's question: *"Mrs. Sara, we know how the bridge will be lifted up, but how will it know there's a flood to raise up?"* This immediately presents a problem that needs a technological solution. The teacher can emphasize the importance of a bridge that automatically lifts during a flood to prevent damage and ensure safety. They can ask a broad opening question like, "Why is it important for some bridges to lift up?" to initiate initial thoughts about the purpose of such a mechanism.

Story Reflection:

- **Questions:**
 - What is the main problem Adam is asking about?

- o Why is it important to detect a flood?
- o Can you think of any ways we currently know if there is a flood happening? (Encourage real-world examples like news reports, weather alerts, visual observation).
- o What kind of device or system do you think could automatically detect a flood? (This encourages initial brainstorming without focusing on specific electrical components yet).
- **Suggested answers:**
 - o Adam wants to know how the bridge will automatically know when there is a flood so it can lift up.
 - o It's important to detect a flood to protect people, property (like the bridge itself), and infrastructure from damage caused by rising water.
 - o We might hear about floods on the news, get weather alerts on our phones, or see the water levels rising ourselves.
 - o Possible initial ideas could include something that senses water level, a sensor that reacts to wetness, or maybe even a camera system. (At this stage, accept a variety of ideas without judgment).
- **Strategy: Think-Pair-Share followed by Whole Class Discussion**
- **Implementation:**
 0. **Think:** After presenting the story and the initial reflection questions, give students a minute or two to think individually about their answers.
 1. **Pair:** Have students pair up with a classmate to discuss their thoughts and ideas with each other. This allows them to articulate their understanding and hear different perspectives.
 2. **Share:** Bring the class back together for a whole-class discussion. The teacher can facilitate this by asking pairs to share their key ideas and answers to the reflection questions. The teacher should acknowledge all contributions and guide the discussion towards the idea that a specific type of sensor or "electrical part" is needed for automatic detection, setting the stage for the next part of the lesson. Mrs. Sara's statement, *"Alright, kids, today we will add a new electrical part that will help detect flooding,"* directly follows this engagement phase and leads into the exploration of that component.

Think

Questions and Possible Answers:

While the image itself doesn't explicitly list questions, based on the instruction to "Discuss which types would be best for our project" after watching a video on flood detection sensors, the "Think" section likely prompts students with questions like:

- What are the different types of flood detection sensors you learned about in the video?
 - o Possible Answers (depending on the video content): Conductive water level sensors, ultrasonic level sensors, float switches, pressure sensors, optical sensors, etc.

- How does each type of sensor work to detect water or water levels?
 - Possible Answers (will vary based on sensor type): Conductive sensors use the conductivity of water to complete a circuit; ultrasonic sensors emit sound waves and measure the time it takes for them to return; float switches use a buoyant object that rises with the water level to trigger a switch; pressure sensors measure the pressure exerted by the water; optical sensors use light beams to detect the presence or level of water.
- What are the advantages and disadvantages of each type of flood detection sensor?
 - Possible Answers (examples):
 - Conductive: Simple, inexpensive, but can corrode and may give false readings with impure water.
 - Ultrasonic: Non-contact, reliable, but can be affected by obstructions or turbulent water.
 - Float switch: Simple, reliable for point-level detection, but mechanical and can get stuck.
 - Pressure: Can measure continuous levels, but needs to be submerged and calibrated.
 - Optical: Accurate, non-contact, but can be affected by dirty water or fog.
- Considering our project (which is implied to be the automatic bridge lifting mechanism discussed in the "Engage" section), which type of flood detection sensor do you think would be the most suitable? Why?
 - Possible Answers (will depend on the specific project requirements, which are not fully detailed in this image): Students might consider factors like the need for continuous level monitoring vs. simple flood/no flood detection, cost, ease of integration with the Arduino, reliability in potentially dirty water conditions, and the range of water level change that needs to be detected. Some might argue for a combination of sensors for redundancy.

Brainstorm Web:

The "Brainstorm Web" would provide a visual space for students to organize their thoughts and make connections between different aspects of flood detection sensors. It might look something like this (with branches extending for each sensor type discussed in the video):

Sensor Type	Working Principle	Advantages	Disadvantages	Potential Project Suitability (Automatic Bridge)
Conductive Water Level Sensors	Utilize the conductivity of water to complete an electrical circuit between probes	Simple design, relatively inexpensive.	Probes can corrode, readings can be affected by impurities in the water, provides	May be suitable for indicating specific critical water levels (e.g., trigger point for bridge lift) but not continuous

	at different heights.		discrete level detection.	monitoring. Requires consideration of water quality.
Ultrasonic Level Sensors	Emit sound waves and measure the time it takes for the echo to return from the water surface.	Non-contact measurement, relatively reliable, can measure continuous levels.	Can be affected by obstructions, turbulent water surfaces, temperature, and humidity. May require calibration.	Potentially good for continuous monitoring of water levels, allowing for proactive bridge adjustments. Need to ensure proper mounting to avoid interference.
Float Switches	Use a buoyant object (float) that rises with the water level to mechanically activate a switch at a specific height.	Simple to understand and implement, reliable for point-level detection.	Mechanical parts can fail or get stuck due to debris, provides only discrete level detection.	Suitable for triggering the bridge lift at a specific water level. Less informative about the rate of rise or continuous level. Requires physical contact with water.
Pressure Sensors (Submersible)	Measure the hydrostatic pressure exerted by the water column above the sensor.	Can provide continuous and accurate level measurements, less affected by surface conditions.	Requires submersion, can be affected by sediment buildup, may need calibration based on water density.	Well-suited for continuous monitoring of water depth, allowing for nuanced responses from the bridge mechanism. Requires careful installation and protection from debris.
Optical Sensors (Infrared/Laser)	Use light beams to detect the presence or level of water based on reflection or refraction.	Accurate, non-contact measurement, can be fast response.	Can be affected by dirty or opaque water, fog, or obstructions in the light path. May be more complex and expensive.	Could be used for precise level detection, but sensitivity to water clarity might be a concern in flood conditions. Requires careful positioning.

Students would fill in the details for each sensor type based on the video and their understanding, eventually leading to a more informed decision about which sensor(s) to use for their automatic bridge project.

Strategies teachers can use to deliver this part:

- **Guided Video Observation:** Before playing the video, provide students with specific questions or aspects to focus on (e.g., types of sensors, how they work, any mentioned advantages or disadvantages). This helps them actively engage with the content.
- **Think-Pair-Share after Video:** After the video, have students individually jot down their initial thoughts on the different sensor types and their potential suitability for the project. Then, have them discuss their ideas with a partner before opening it up to a whole-class discussion.
- **Categorization Activity:** Provide students with a list of sensor characteristics (e.g., cost, accuracy, contact/non-contact, complexity) and have them categorize the different sensor types based on these characteristics. This helps in comparative analysis.
- **Pros and Cons Chart:** As a class or in groups, create a table or chart listing each sensor type and its advantages and disadvantages. This structured approach aids in evaluating the trade-offs of each option.
- **"Which Sensor Would You Choose?" Scenario:** Present different scenarios related to the bridge project (e.g., cost limitations, need for high accuracy, potential for debris in the water) and ask students to justify which sensor type would be the best fit for each scenario. This encourages critical thinking and application of their understanding.
- **Jigsaw Activity:** Divide the class into groups, with each group focusing on one or two types of sensors from the video. Each group researches and prepares a short presentation on their assigned sensors, highlighting their working principle, pros, and cons. Then, the groups share their findings with the rest of the class, creating a comprehensive understanding of all the sensor options. This promotes collaborative learning and peer teaching.

2.Explore

Teacher Role:

- Ensuring students have the necessary materials and understand how to handle them safely.

- Guiding students through the wiring of the sensor to the Arduino, potentially providing diagrams or demonstrations.
- Assisting students in opening and using the Arduino serial monitor to view the sensor readings.
- Encouraging students to make observations and record their findings as they change the water level.
- Facilitating discussion and helping students interpret the data they collect.
- Answering questions and troubleshooting any issues that may arise during the activity.

Expected From Students:

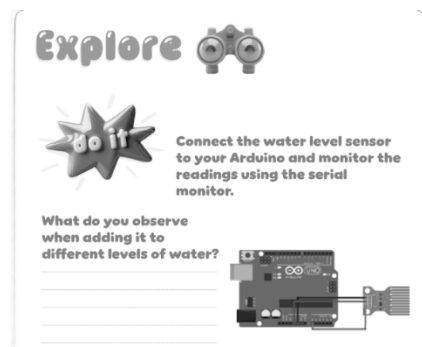
- Carefully connecting the water level sensor to the Arduino board following instructions or guidance.
- Opening and monitoring the serial monitor on their computer to observe the sensor readings.
- Actively experimenting by immersing the sensor in water at different levels.
- Observing and recording the changes in the serial monitor readings as the water level changes.
- Attempting to identify any patterns or relationships between the water level and the sensor readings.
- Reflecting on their observations and formulating answers to the reflection questions.

Parts of the books included:

- **Activity - Do It:**

- **Materials Needed:**

- Arduino board (e.g., Uno)
- Water level sensor (the specific type is not detailed in this image, but it could be resistive, capacitive, or another type)



- Jumper wires (male-to-male)
- A container to hold water (e.g., a cup or beaker)
- Water
- USB cable to connect the Arduino to a computer
- Computer with the Arduino IDE installed

Step-by-Step Instructions

- Connect the sensor to the Arduino: Refer to the datasheet or instructions for your specific water level sensor to identify the function of each pin (e.g., VCC, GND, Signal). Use jumper wires to connect these pins to the appropriate pins on the Arduino (e.g., 5V or 3.3V for VCC, GND for ground, and an analog input pin like A0 for the signal).
- Open the Arduino IDE: Launch the Arduino software on your computer.
- Write and upload the code: Write a simple Arduino sketch that reads the analog value from the connected sensor pin and prints it to the serial monitor. For an analog sensor, this might involve using the `analogRead()` function. Ensure you select the correct board and port in the Arduino IDE and upload the code to your Arduino.
- Open the Serial Monitor: In the Arduino IDE, open the Serial Monitor (usually found under the "Tools" menu). You should see the sensor readings being printed.
- Experiment with water levels: Carefully place the water level sensor into the container with water, starting with a low water level. Observe and record the readings in the serial monitor.
- Increase the water level gradually: Add more water to the container in increments and observe how the readings in the serial monitor change at each level. Record your observations.
- Repeat: Continue adding water and observing the readings across a range of water levels.
- **Reflection Questions:**
 - What do you observe in the serial monitor readings when the water level is low?
 - How do the readings change as you add more water and the water level increases?

- Is there a consistent pattern or relationship between the water level and the sensor readings? Describe it.
- What do you think the range of values you are seeing in the serial monitor represents in terms of water level?
- What are some potential challenges or limitations you might encounter when using this type of sensor to detect flood levels in a real-world scenario?

3.Explain

Teacher Role:

- Introducing the purpose of learning about water level sensors in the context of the automatic bridge project.
- Guiding students to focus on key information during the video.
- Asking probing questions to encourage critical thinking and analysis of the video content.
- Clarifying any misconceptions or confusing points.
- Facilitating a discussion where students share their observations and insights from the video.
- Introducing and guiding students through any provided reading material on water level sensors.
- Helping students connect the information from the video and reading to their hands-on exploration in the previous section and the overall project goal.

Expected From Students:

Students are expected to **engage actively** with the video and any reading material, **share their observations** and understanding during discussions, **ask clarifying questions**, and **connect the explained concepts** to their prior experiences and the project.

Parts of the books included:



Watch It:

- **Goal:** Learn more about water level sensors by watching the provided video.
- **Pre-Watching: Set a purpose.**
 - **Strategy: Anticipation Guide**
 - **Implementation:** Before showing the video, the teacher can present 2-3 statements related to water level sensors (some true, some potentially false or requiring more information). Students individually mark whether they agree or disagree with each statement. After watching the video, they revisit their initial responses and revise them based on the information they learned. This sets a purpose for watching and encourages active information gathering to confirm or refute their initial understanding.
Example statements:
 - "All water level sensors work by physically touching the water." (Likely false)
 - "Water level sensors can be used to prevent flooding." (Potentially true, depending on the application)
 - "There is only one type of water level sensor." (Likely false)
- **During Watching: Active engagement.**
 - **Strategy: Note-Taking with Prompts**
 - **Implementation:** The teacher can provide students with a simple note-taking template or guide with specific prompts to focus their attention while watching the video. Examples of prompts:

- "List the different types of water level sensors mentioned."
- "How does [mention a specific sensor type] work?"
- "What are some advantages or disadvantages of [mention a specific sensor type]?"
- "What are some real-world applications of water level sensors shown in the video?" Students actively fill in the template as they watch, ensuring they are looking for specific information.
- **Post-Watching: Discussion and reflection.**
 - **Strategy: Think-Pair-Share followed by Class Brainstorm**
 - **Implementation:** After the video, students first individually reflect on the video content and their notes. Then, they pair up with a classmate to discuss their key takeaways, any surprising information, and answers to the pre-watching anticipation guide or during-watching prompts. Finally, the teacher facilitates a whole-class discussion, encouraging pairs to share their insights and creating a class brainstorm on the different types of water level sensors, their working principles, advantages, disadvantages, and potential applications, especially in the context of the automatic bridge.
 - about water level sensors. Then, the teacher guides them to brainstorm "W" (What We Want to Know) by asking questions like: "What aspects of water level sensors were not fully explained in the video?", "What other details would be helpful for our project?", or "Are there different ways these sensors are used in different situations?". This activates prior knowledge and sets a purpose for reading.

4.Elaborate

Teacher Role:

- The teacher's role in this "Elaborate" phase is to **encourage creativity and the application of knowledge** by guiding students to build a circuit and write code that uses the water level sensor to control multiple LEDs.
- They should encourage students to think about how different sensor readings can trigger different outputs.

Expected From Students:

- Students are expected to **create models, circuits, experiments, or code** to make different LEDs light up based on varying water levels detected by the sensor.
- This demonstrates their ability to integrate sensor input with output control using the Arduino.

Parts of the books included:

Assessment:

Assessment

Focus



What is the primary function of a water level sensor in an Arduino project?

- A) To measure the temperature of the water.
- B) To detect the height or level of water in a container.**
- C) To control the speed of water flow.
- D) To measure the pH of the water.

Where can we connect signal pin of water level sensor at Arduino?

- A) Analog pins.**
- B) digital pins.
- C) power pins.

Strategies to apply while answering:

1. **Recall the name:** The name "water level sensor" itself strongly suggests its function is related to the level of water.
2. **Consider the context:** The preceding activities focused on exploring and explaining how these sensors respond to changes in water height.

3. **Eliminate incorrect options:**

- Temperature sensors measure heat.
- Flow sensors measure the rate of fluid movement.
- pH sensors measure the acidity or alkalinity of a liquid.

4. **Recall the "Explore" activity:** In the "Explore" section, students likely connected the signal pin to an analog input to read a range of values corresponding to different water levels.

5. **Consider the nature of the data:** Water level sensors typically output a variable voltage or resistance that changes with the water level, requiring an analog-to-digital conversion, which is done by the Arduino's analog input pins.

6. **Eliminate incorrect options:**

- Digital pins read discrete HIGH or LOW states, not a continuous range of values. While some advanced water level sensors might have digital outputs for specific thresholds, the primary signal for level sensing is usually analog.
- Power pins (like 5V or 3.3V) are used to supply power to components, not to receive signal inputs.

• **Practice.**

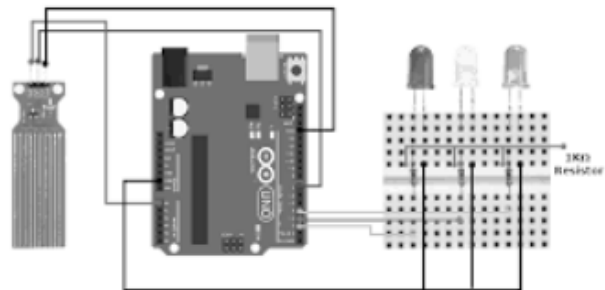
- **Goal:** Let's make different LEDs light up as the water level increases. Build this circuit and write the code for it.
- **Materials Needed (based on the image):**
 - Arduino board (Uno or similar)
 - Water level sensor
 - Three LEDs (different colors are helpful, e.g., red, yellow, green)

- Three current-limiting resistors (typically 220Ω)
 - Breadboard
 - Jumper wires
 - Container for water
 - Water
 - USB cable for Arduino
- **Step-by-Step Instructions :**
 0. **Connect the water level sensor to the Arduino:** Connect the VCC and GND of the sensor to the 5V and GND of the Arduino, respectively. Connect the signal pin of the sensor to an analog input pin on the Arduino (e.g., A0).
 1. **Connect the LEDs with resistors:** For each LED, connect the longer leg (anode) to a digital output pin on the Arduino (e.g., pins 9, 10, 11) through a current-limiting resistor. Connect the shorter leg (cathode) of each LED to the GND rail on the breadboard, which is then connected to the Arduino's GND.
 2. **Build the circuit on the breadboard:** Arrange the components and wires according to a circuit diagram (similar to the one shown in the image, but adapted for the specific water level sensor).

Practice



Let's make different LEDs light up as the water level increases. Build this circuit and write the code for it."



3. Write the Arduino code:

- Define variables for the analog input pin connected to the sensor and the digital output pins connected to the LEDs.
- In the setup() function, set the LED pins as OUTPUTs and initialize serial communication for debugging.
- In the loop() function:
 - Read the analog value from the water level sensor using analogRead().
 - Map the range of analog values (which will correspond to different water levels) to different conditions for the LEDs. For example:
 - If the analog reading is below a certain threshold (low water level), turn on the red LED.
 - If the analog reading is within a middle range (medium water level), turn on the yellow LED.
 - If the analog reading is above a certain threshold (high water level), turn on the green LED.
- Use digitalWrite() to control the state (HIGH/LOW) of the LED pins based on the water level.
- Include a small delay() for stability.

```
// Define the analog pin for the water level sensor
```

```
const int waterLevelPin = A0;
```



```
// Define the digital pins for the LEDs

const int redLedPin = 9;
const int yellowLedPin = 10;
const int greenLedPin = 11;


// Define thresholds for water levels (adjust these based on your sensor's readings)
int lowWaterThreshold = 200; // Example analog reading for low water
int mediumWaterThreshold = 500; // Example analog reading for medium water
int highWaterThreshold = 800; // Example analog reading for high water


void setup() {
    // Set the LED pins as outputs
    pinMode(redLedPin, OUTPUT);
    pinMode(yellowLedPin, OUTPUT);
    pinMode(greenLedPin, OUTPUT);

    // Initialize serial communication for debugging
    Serial.begin(9600);
}


void loop() {
    // Read the analog value from the water level sensor
    int sensorValue = analogRead(waterLevelPin);

    Serial.print("Sensor Value: ");
    Serial.println(sensorValue);
}
```

```
// Control the LEDs based on the water level

if (sensorValue < lowWaterThreshold) {

    digitalWrite(redLedPin, HIGH);

    digitalWrite(yellowLedPin, LOW);

    digitalWrite(greenLedPin, LOW);

} else if (sensorValue >= lowWaterThreshold && sensorValue <
mediumWaterThreshold) {

    digitalWrite(redLedPin, LOW);

    digitalWrite(yellowLedPin, HIGH);

    digitalWrite(greenLedPin, LOW);

} else if (sensorValue >= mediumWaterThreshold && sensorValue <
highWaterThreshold) {

    digitalWrite(redLedPin, LOW);

    digitalWrite(yellowLedPin, LOW);

    digitalWrite(greenLedPin, HIGH);

} else { // Very high water level

    // You can define behavior for very high water if needed

    digitalWrite(redLedPin, LOW);

    digitalWrite(yellowLedPin, LOW);

    digitalWrite(greenLedPin, HIGH); // Or blink all LEDs, etc.

}

// Delay for a short time

delay(100);

}
```

4. **Upload the code to the Arduino:** Connect the Arduino to your computer via USB and upload the written code using the Arduino IDE.
5. **Test the setup:** Gradually add water to the container with the water level sensor. Observe how different LEDs light up as the water level changes, corresponding to the thresholds defined in your code.

○ **Reflection Questions:**

0. How did you determine the analog value ranges that correspond to different water levels?
1. What changes would you need to make to the code if you wanted the LEDs to fade in and out instead of just turning on and off?
2. How could you calibrate the sensor to provide more accurate water level readings?
3. What other ways could you represent the water level using different output components connected to the Arduino?
4. How could this basic principle be expanded to create a more sophisticated flood warning system?

Now go to the plan part in your EDP paper and add this enhancement



5. Evaluate

Teacher Role:

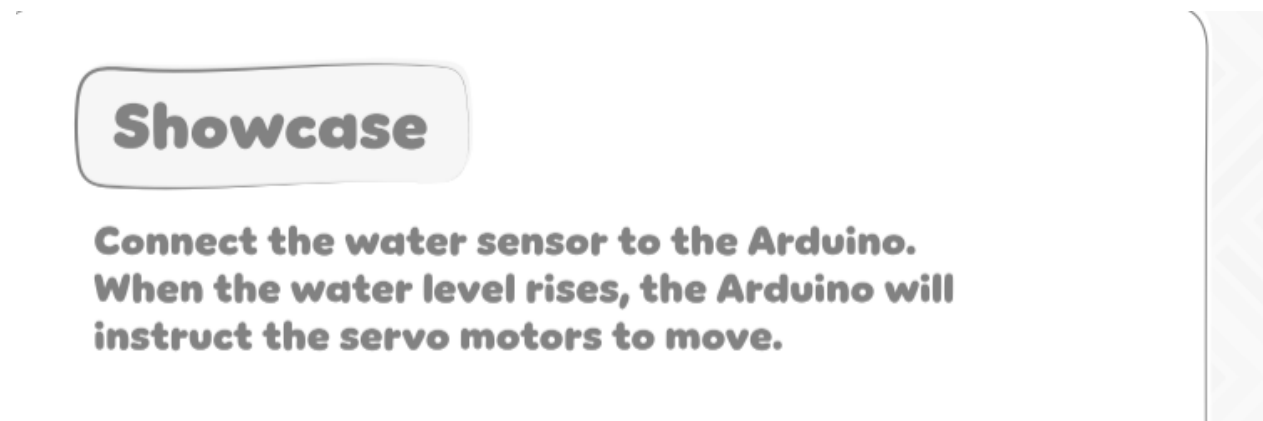
- Observing students' work during the "Showcase."

- Using a rubric (provided below) to assess different aspects of their project.
- Asking students questions about their design choices, code, and the functionality of their project.
- Providing constructive feedback to help students identify areas of strength and areas for improvement.
- Connecting the project outcomes to the learning objectives of the module.

Expected From Students:

Students are expected to **answer questions** about their project, **demonstrate their understanding** of how the water level sensor and servo motors work together, and **present their working project** as the "Showcase."

Parts of the books included:



Showcase: Evaluate the students' work in the main project and add a rubric so teachers can follow.

The main project described in the "Showcase" is: **Connect the water sensor to the Arduino. When the water level rises, the Arduino will instruct the servo motors to move.** This implies a system where the water level sensor detects a change, and this change triggers a specific movement in one or more servo motors (likely mimicking the lifting of a bridge).

Rubric for Evaluating the Automatic Bridge Project:

Criteria	Excellent (4 Points)	Proficient (3 Points)	Developing (2 Points)	Needs Improvement (1 Point)
Circuit Construction	The water level sensor and servo motor(s) are correctly wired to the Arduino with appropriate power and signal connections. The circuit is neat, organized, and functions reliably.	The water level sensor and servo motor(s) are mostly correctly wired, and the circuit generally functions. There might be some minor wiring issues or lack of organization.	There are significant errors in the wiring of the water level sensor and/or servo motor(s). The circuit may function inconsistently or not at all.	The wiring is incorrect or incomplete, preventing the circuit from functioning.
Code Functionality	The Arduino code accurately reads the water level sensor and correctly controls the servo motor(s) to move in a logical way based on changes in water level. The code is well-commented and easy to understand.	The Arduino code reads the water level sensor and controls the servo motor(s), but there might be minor issues in the responsiveness or the logic of the movement based on water level. The code has some comments.	The Arduino code attempts to read the sensor and control the servo motor(s), but there are significant errors in the logic or implementation. The servo movement may not be related to the water level.	The Arduino code is incomplete, contains major errors, or does not attempt to read the sensor and control the servo motor(s) appropriately.
Understanding of Concepts	The student demonstrates a clear understanding of how the water level sensor	The student demonstrates a general understanding of the core concepts but	The student shows a limited understanding of how the sensor and servo motors work together or	The student demonstrates a lack of understanding of the fundamental

	works, how servo motors are controlled, and how the Arduino acts as the interface between them. They can explain their code and circuit choices effectively.	might have some minor inaccuracies or struggle to fully explain all aspects of their project.	how the Arduino controls them. Their explanations may be unclear or inaccurate.	concepts related to the water level sensor, servo motors, and Arduino control.
Problem Solving	The student effectively troubleshoots any issues that arise during the project and demonstrates resourcefulness in finding solutions.	The student identifies and resolves some issues with the project, possibly with some teacher guidance.	The student struggles to identify and resolve issues with the project and relies heavily on teacher assistance.	The student is unable to identify or resolve problems preventing the project from functioning.
Project Presentation	The student clearly and effectively presents their project, explaining its functionality and the relationship between the water level and the servo motor movement.	The student presents their project adequately, explaining its basic functionality.	The student struggles to clearly present their project or explain how it works.	The student is unable to present their project or explain its functionality.

vRelation to Standards:

This project and its evaluation can be linked to various educational standards depending on the curriculum. Some potential connections include:

- **Science and Technology Standards:** Understanding sensors, actuators, feedback loops, and the application of technology to solve real-world problems (like flood detection and mitigation).
- **Engineering Design Standards:** Following a design process (conceive, design, implement, operate), troubleshooting, and evaluating the effectiveness of a solution.
- **Computer Science Standards:** Programming microcontrollers, understanding input and output, conditional logic, and potentially data processing.
- **Mathematics Standards:** Understanding data representation and potentially using mathematical relationships to map sensor readings to servo motor movement.

UNIT 03 [Advanced]

↳ **Chapter 01 : Robotics.**

LESSON 01

Show Your Results.

Time Frame : 90 mins

Objectives:

- Identify LCD and i2c module.
- Understand how to connect the LCD with our project.

SEL Relation:

Challenge questions in this lesson:

1.Engage

Teacher Role:

The teacher's responsibilities in this "Engage" phase include **facilitating a discussion** that builds upon the previous lesson about the automatic bridge and **asking questions** to guide students towards considering how to communicate flood warnings to citizens. The teacher should encourage creative thinking and problem-solving.

Expected From Students:

Students are expected to be **participating in a discussion** about how to enhance their bridge project, **sharing their ideas** for a flood warning system, and **responding to the teacher's questions** with thoughtful suggestions.

Parts of the books included:

Intro Story:



- **Strategy: Building on Prior Knowledge (Review and Expansion)**
- **Implementation:** The teacher can start by referencing the successful construction of the automatic bridge from the previous lesson, as stated by Mrs. Sara: *"Alright, team! We've built our bridge."* This immediately connects to their prior work and creates a sense of accomplishment. Then, the teacher introduces a new layer to the project by posing the central question: *"Now, let's think about what we can add to let the citizens know if there is a flood."* This presents a practical problem that requires a solution, encouraging students to think about the user experience

and the importance of communication in a real-world application. The teacher can ask an open-ended question like, "Why is it important for people to know if there's a flood?" to initiate the discussion.

Story Reflection:

- **Questions:**

- What is the new problem Mrs. Sara wants us to think about?
- Why is it not enough for just the bridge to lift automatically?
- What are some ways we currently receive warnings about dangerous situations? (Encourage brainstorming of various methods like sirens, announcements, phone alerts, visual signs).
- Laila suggests using a screen. What kind of information could we display on a screen to warn people about a flood?

- **Suggested answers:**

- Mrs. Sara wants us to figure out how to inform people that there is a flood.
- People need to know about the flood so they can take precautions, avoid the area, and stay safe. The bridge lifting alone doesn't directly communicate this danger.
- We receive warnings through sirens, public announcements, news broadcasts, weather apps on our phones, road signs, etc.
- On a screen, we could display a warning message like "FLOOD WARNING!", show the current water level, indicate if the bridge is about to lift or is lifted, or provide instructions for citizens.

- **Strategy: Think-Pair-Share followed by Whole Class Idea Generation**

- **Implementation:**

0. **Think:** After presenting the story and the initial reflection questions, give students a minute to individually consider their ideas for a flood warning system, focusing on how to communicate the information visually.
1. **Pair:** Have students pair up to discuss their individual ideas and build upon each other's suggestions for how to warn citizens using a screen or other methods.

2. **Share:** Bring the class back together for a whole-class discussion. The teacher can record the various ideas generated by the students on a whiteboard or shared document. This allows for a collective brainstorming session, highlighting the potential of using a screen (as suggested by Laila) and other possibilities, setting the stage for introducing the LCD and i2c module as a potential solution for this visual communication.

Think

Based on the ongoing scenario of an automatic bridge and the need for a flood warning system, the "Think" section should prompt students to consider *how* the system knows a flood is occurring in the first place.

Questions and Possible Answers:

- How does our automatic bridge system know when there is a flood?
 - Possible Answers: It uses a water level sensor. The sensor detects the height of the water. When the water reaches a certain level, the sensor sends a signal.
- What kind of information does the water level sensor provide to the Arduino?
 - Possible Answers: It provides a reading, likely a numerical value. This value changes as the water level changes.
- How does the Arduino decide if the water level indicates a flood?
 - Possible Answers: The Arduino code is programmed with specific threshold values. When the sensor reading goes above a certain threshold, the Arduino interprets it as a flood.
- What are some different ways a water level sensor might work to give us a reading? (Connect back to previous lessons on sensor types if applicable).
 - Possible Answers: It might measure how much of a probe is covered in water, it might send out sound waves and measure the distance to the water, it might measure pressure.

Brainstorm Web:

The "Brainstorm Web" would focus on the process of flood detection within the system:

Think

Based on the ongoing scenario of an automatic bridge and the need for a flood warning system, the "Think" section should prompt students to consider *how* the system knows a flood is occurring in the first place.

Questions and Possible Answers:

- How does our automatic bridge system know when there is a flood?
 - Possible Answers: It uses a water level sensor. The sensor detects the height of the water. When the water reaches a certain level, the sensor sends a signal.
- What kind of information does the water level sensor provide to the Arduino?
 - Possible Answers: It provides a reading, likely a numerical value. This value changes as the water level changes.
- How does the Arduino decide if the water level indicates a flood?
 - Possible Answers: The Arduino code is programmed with specific threshold values. When the sensor reading goes above a certain threshold, the Arduino interprets it as a flood.
- What are some different ways a water level sensor might work to give us a reading? (Connect back to previous lessons on sensor types if applicable).
 - Possible Answers: It might measure how much of a probe is covered in water, it might send out sound waves and measure the distance to the water, it might measure pressure.

Brainstorm Web:

The "Brainstorm Web" would focus on the process of flood detection within the system:

How We Know Flooding is Happening

Think

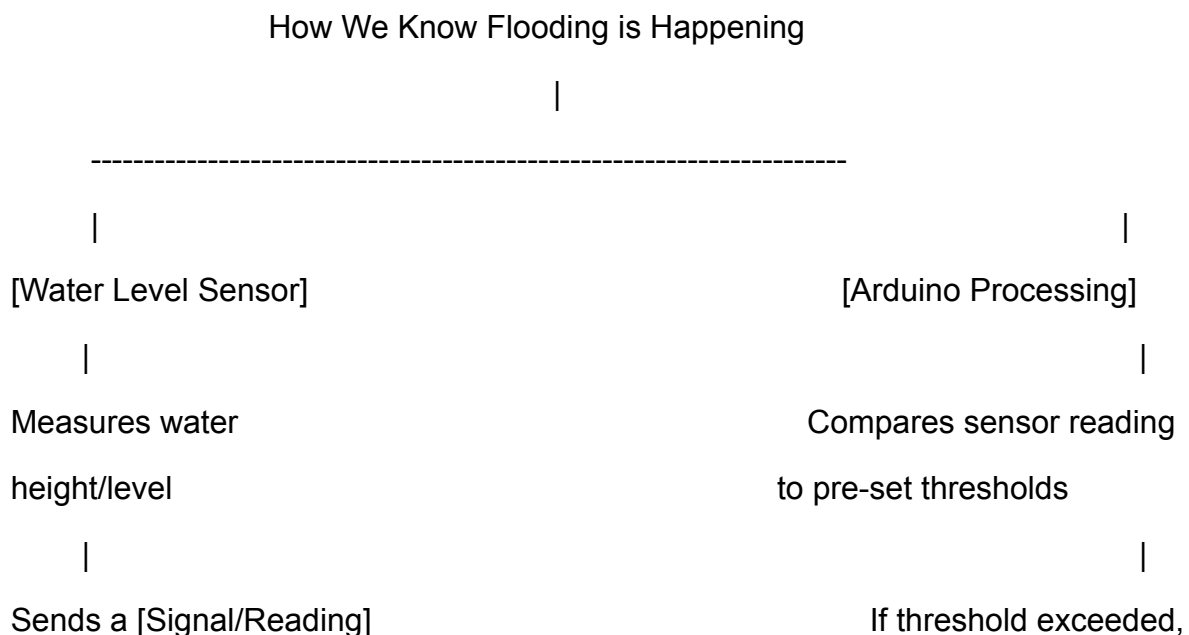
Based on the ongoing scenario of an automatic bridge and the need for a flood warning system, the "Think" section should prompt students to consider *how* the system knows a flood is occurring in the first place.

Questions and Possible Answers:

- How does our automatic bridge system know when there is a flood?
 - Possible Answers: It uses a water level sensor. The sensor detects the height of the water. When the water reaches a certain level, the sensor sends a signal.
- What kind of information does the water level sensor provide to the Arduino?
 - Possible Answers: It provides a reading, likely a numerical value. This value changes as the water level changes.
- How does the Arduino decide if the water level indicates a flood?
 - Possible Answers: The Arduino code is programmed with specific threshold values. When the sensor reading goes above a certain threshold, the Arduino interprets it as a flood.
- What are some different ways a water level sensor might work to give us a reading? (Connect back to previous lessons on sensor types if applicable).
 - Possible Answers: It might measure how much of a probe is covered in water, it might send out sound waves and measure the distance to the water, it might measure pressure.

Brainstorm Web:

The "Brainstorm Web" would focus on the process of flood detection within the system:



(Analog/Digital Value)

interprets as "Flood"

Strategies teachers can use to deliver this part:

- **Review and Connect:** Begin by briefly reviewing the previous lesson on the automatic bridge and the role of the water level sensor in triggering the bridge mechanism. Ask students to recall how the sensor worked in their previous experiments.
- **Think Aloud:** The teacher can model the thinking process by saying, "We know the bridge lifts when there's a flood, but how does the system *know* it's a flood? It doesn't have eyes to see the water rising. It must rely on the..." (pause for student input - "water level sensor"). "And what kind of information does that sensor give the Arduino?"
- **"If-Then" Scenarios:** Present simple "if-then" scenarios to illustrate the threshold concept: "If the sensor reading is below 300, is it a flood? If it's above 700, is it a flood? What happens in between?" This helps students understand how the Arduino makes a decision based on the sensor data.
- **Visual Representation:** Use a simple graph or diagram to illustrate how the sensor reading changes with the water level and how the threshold values are used to define what constitutes a "flood" condition. This can make the abstract concept of data interpretation more concrete.

2,Explore

Teacher Role:

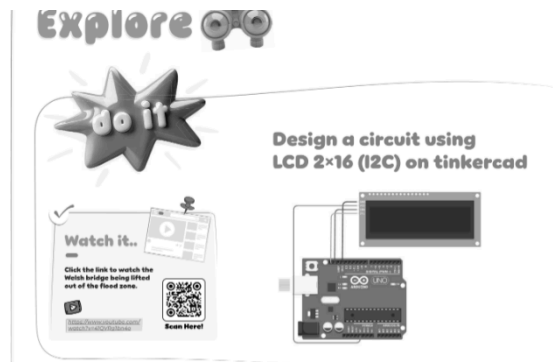
- Ensuring students have access to Tinkercad and understand how to navigate the Circuits environment.
- Guiding students on how to find the necessary components (Arduino Uno, LCD 2x16, I2C module) in the Tinkercad parts library.
- Assisting students in understanding the basic wiring connections between the Arduino, the I2C module, and the LCD.

- Encouraging students to experiment with the Tinkercad code editor to display text on the LCD.
- Answering questions and troubleshooting any issues students encounter while designing their circuit and code.

Expected From Students:

- Navigating Tinkercad and locating the Arduino Uno, LCD 2x16, and I2C module components.
- Designing and building the virtual circuit in Tinkercad, connecting the components according to the basic principles of I2C communication.
- Exploring the Tinkercad code editor and attempting to write basic Arduino code to display text on the LCD.
- Observing the output on the virtual LCD screen in the Tinkercad simulation.
- Reflecting on the process and formulating answers to the reflection questions.

Parts of the books included:



Activity - Do It:

- **Materials Needed (virtual, within Tinkercad):**
 - Arduino Uno board
 - LCD 2x16 display
 - I2C module (specifically designed for 16x2 LCDs, often labeled as "I2C LCD")
 - Jumper wires (virtual)

- Computer with internet access and a Tinkercad account.
- **Step-by-Step Instructions (within Tinkercad):**
 0. **Create a new Tinkercad Circuit:** Open Tinkercad and create a new circuit design.
 1. **Add Components:** Search for and add an "Arduino Uno," an "LCD 16x2," and an "I2C LCD" module to your virtual breadboard area.
 2. **Connect the I2C Module to the LCD:** The I2C module typically plugs directly into the back of the 16x2 LCD. In Tinkercad, you will virtually connect the corresponding pins. Look for labels like VCC, GND, SDA, and SCL on both the LCD/I2C module and the Arduino.
 3. **Connect the I2C Module to the Arduino:**
 - Connect the **VCC** pin of the I2C module to the **5V** pin on the Arduino.
 - Connect the **GND** pin of the I2C module to the **GND** pin on the Arduino.
 - Connect the **SDA** (Serial Data) pin of the I2C module to the **SDA (A4)** pin on the Arduino Uno.
 - Connect the **SCL** (Serial Clock) pin of the I2C module to the **SCL (A5)** pin on the Arduino Uno.
 4. **Open the Code Editor:** Click on the "Code" button in Tinkercad.
 5. **Include the LiquidCrystal I2C Library:** Tinkercad has a built-in library for I2C LCDs. In the code editor, go to "Libraries" and include the "LiquidCrystal I2C" library.
 6. **Write Basic Code:** Write code to initialize the LCD and display some text. You will likely need to specify the I2C address of your LCD module (common addresses are 0x27 or 0x3F; you might need to try both if the first doesn't work). A basic code structure might look like this:

```
#include <LiquidCrystal_I2C.h>
```



```
// Set the LCD address to 0x27 for a 16 chars and 2 line display
LiquidCrystal_I2C lcd(0x27, 16, 2); // Adjust address if needed
```

```
void setup() {
  lcd.init();    // Initialize the LCD
  lcd.backlight(); // Turn on the backlight
  lcd.print("Hello, world!");
  lcd.setCursor(0, 1); // Set cursor to the second line (row 1)
  lcd.print("Tinkercad I2C");
}
```

```
void loop() {
  // No looping action needed for this basic display
}
```

7. **Start Simulation:** Click the "Start Simulation" button to see if your circuit and code work. The text "Hello, world!" should appear on the virtual LCD. If it doesn't, check your wiring and try a different I2C address in the code.

- **Reflection Questions:**

0. What are the advantages of using an I2C module with the LCD compared to wiring it directly to the Arduino? (Hint: Think about the number of wires).
1. What does the I2C address of the LCD refer to? How might you find the correct address if the default doesn't work? (In a real-world scenario, you might need to use an I2C scanner sketch).
2. How would you display different messages on the two lines of the LCD?
3. How could you use the `lcd.setCursor()` function to position text at different locations on the screen?
4. Can you think of ways you could use an LCD screen to provide feedback or display information in your automatic bridge project?

4.Elaborate

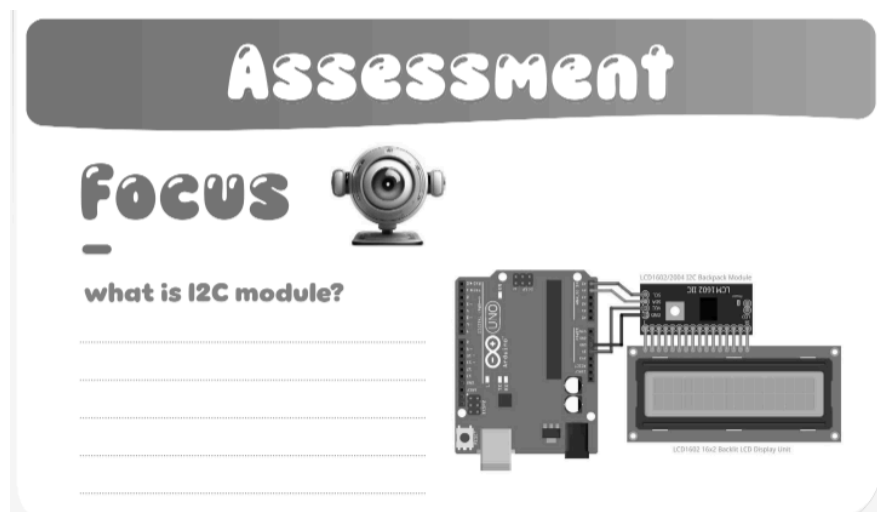
Teacher Role:

- The teacher's role in this "Elaborate" phase is to **encourage creativity and application of knowledge** by guiding students to connect LEDs to their existing water level sensor system.
- They should prompt students to think about how to map sensor readings to different LED states and reinforce the concept of using outputs to visualize sensor data.

Expected From Students:

Students are expected to **create models, circuits, experiments, or code** that extend their water level monitoring system to include three LEDs that illuminate based on different water level ranges detected by the sensor.

Parts of the books included:



Assessment:

- **Focus:**
 - **Question:** What is an I2C module?
 - **Answer:** An I2C (Inter-Integrated Circuit) module is a small electronic circuit that allows microcontrollers, like the Arduino, to communicate with other devices using only two wires (SDA - Serial Data and SCL - Serial

Clock) plus power and ground. It simplifies the wiring needed to interface with multiple peripherals, such as LCD displays, sensors, and memory devices. In the context of an LCD, an I2C module is often attached to the back of the LCD to reduce the number of Arduino pins required to control it.

- **Strategies to apply while answering:**
 1. **Recall the "Explore" activity:** Remember using the I2C module with the LCD, noting the reduced number of connections to the Arduino.
 2. **Consider the problem it solves:** Think about why using many wires for each peripheral would be inefficient and how I2C addresses this.
 3. **Identify key characteristics:** Focus on the two data/clock wires and the concept of addressing multiple devices.
- **Practice: Enhance our water level monitoring system by adding three LEDs display.**
 - **Goal:** Integrate three LEDs into the water level monitoring system to provide a visual indication of low, medium, and high water levels.
 - **Materials Needed:**
 - Existing Arduino setup with a water level sensor connected.
 - Three LEDs (different colors recommended: e.g., red, yellow, green).
 - Three current-limiting resistors (typically 220Ω for standard LEDs).
 - Breadboard and jumper wires.
 - **Step-by-Step Instructions:**
 - **Connect the LEDs:**
 - Connect the longer leg (anode) of each LED to a separate digital output pin on the Arduino (e.g., pins 9, 10, and 11) through a current-limiting resistor. Connect one end of the

resistor to the LED's anode and the other end to the Arduino's digital pin.

- Connect the shorter leg (cathode) of each LED to the GND rail on the breadboard, which is connected to the Arduino's GND.
- **Modify the Arduino Code:**
 - **Define LED Pins:** At the beginning of your Arduino sketch, define integer variables for the digital pins connected to each LED (e.g., `const int redLedPin = 9;`, `const int yellowLedPin = 10;`, `const int greenLedPin = 11;`).
 - **Set Pin Modes:** In the `setup()` function, set these LED pins as OUTPUT using `pinMode()`.
 - **Read Sensor Value:** In the `loop()` function, read the analog value from your water level sensor (as you were doing previously).
 - **Establish Thresholds:** Determine appropriate threshold values for your water level sensor that correspond to low, medium, and high water levels. You might need to experiment with the sensor readings to find suitable thresholds.
 - **Control LEDs based on Thresholds:** Use `if`, `else if`, and `else` statements to control the LEDs based on the sensor value:
 - If the sensor value is below the "low" threshold, turn on the red LED and turn off the others.
 - If the sensor value is between the "low" and "medium" thresholds, turn on the yellow LED and turn off the others.

- If the sensor value is above the "medium" threshold (and potentially below a "high" threshold, if you define one), turn on the green LED and turn off the others.
- Use digitalWrite(ledPin, HIGH) to turn an LED on and digitalWrite(ledPin, LOW) to turn it off.

```
// Define the analog pin for the water level sensor
```

```
const int waterLevelPin = A0;
```

```
// Define the digital pins for the LEDs
```

```
const int redLedPin = 9;
```

```
const int yellowLedPin = 10;
```

```
const int greenLedPin = 11;
```

```
// Define thresholds for water levels (adjust these based on your sensor's readings)
```

```
int lowWaterThreshold = 300; // Example analog reading for low water
```

```
int mediumWaterThreshold = 600; // Example analog reading for medium water
```

```
void setup() {
```

```
  // Set the LED pins as outputs
```

```
  pinMode(redLedPin, OUTPUT);
```

```
  pinMode(yellowLedPin, OUTPUT);
```

```
  pinMode(greenLedPin, OUTPUT);
```

```
  // Initialize serial communication for debugging
```

```
  Serial.begin(9600);
```

```
}
```

```
void loop() {  
    // Read the analog value from the water level sensor  
    int sensorValue = analogRead(waterLevelPin);  
    Serial.print("Sensor Value: ");  
    Serial.println(sensorValue);  
  
    // Control the LEDs based on the water level  
    if (sensorValue < lowWaterThreshold) {  
        digitalWrite(redLedPin, HIGH);  
        digitalWrite(yellowLedPin, LOW);  
        digitalWrite(greenLedPin, LOW);  
    } else if (sensorValue >= lowWaterThreshold && sensorValue <  
mediumWaterThreshold) {  
        digitalWrite(redLedPin, LOW);  
        digitalWrite(yellowLedPin, HIGH);  
        digitalWrite(greenLedPin, LOW);  
    } else if (sensorValue >= mediumWaterThreshold) {  
        digitalWrite(redLedPin, LOW);  
        digitalWrite(yellowLedPin, LOW);  
        digitalWrite(greenLedPin, HIGH);  
    }  
  
    // Delay for a short time  
    delay(100);  
}
```

- **Upload and Test:** Upload the modified code to your Arduino.
Observe how the LEDs change state as you vary the water level detected by the sensor.
- **Reflection Questions:**
 - How did you determine the appropriate threshold values for the low, medium, and high water levels based on your sensor's readings?
 - What are the advantages of using LEDs to provide a quick visual indication of the water level?
 - How could you calibrate your system to ensure the LEDs accurately represent the water levels?
 - Could you expand this system to use more LEDs to provide a more granular indication of the water level? How would the code need to change?
 - How does this LED display complement the idea of using an LCD screen for displaying more detailed information?

5.Evaluate

Teacher Role:

- Observing students' work during the "Showcase."
- Using a rubric (to be developed) to assess different aspects of their project.
- Asking students questions about their design choices, code, and the functionality of their smart bridge.
- Providing constructive feedback to help students identify areas of strength and areas for improvement.
- Reviewing the "Now I can..." self-assessment to gauge student confidence and understanding.

- Connecting the project outcomes to the learning objectives of the module.

Expected From Students:

Students are expected to **demonstrate their understanding** by presenting their working smart bridge, **explain their design and code**, and **answer questions** about how the different components interact. They are also expected to **reflect on their learning** by completing the "Now I can..." statements.

Parts of the books included:



Let's make our smart bridge using 2 servo motors, water level sensor and LCD.

Showcase: Evaluate the students' work in the main project and add a rubric so teachers can follow.

The main project described in the "Showcase" is: **Let's make our smart bridge using 2 servo motors, water level sensor and LCD.** This implies a system where the water level sensor detects a change, the Arduino controls the two servo motors to simulate the bridge lifting, and the LCD displays relevant information (e.g., water level status, warning messages).

Rubric for Evaluating the Smart Bridge Project:

Criteria	Excellent (4 Points)	Proficient (3 Points)	Developing (2 Points)	Needs Improvement (1 Point)
Circuit Construction	All components (water level sensor, 2 servo motors, LCD with I2C) are correctly wired to the Arduino. The	All components are mostly correctly wired, and the circuit generally functions. There might be some	There are significant errors in the wiring, and the circuit functions inconsistently or not at all.	The wiring is incorrect or incomplete, preventing the circuit from functioning.

	circuit is neat, organized, and functions reliably.	minor wiring issues or lack of organization.		
Code Functionality	The Arduino code accurately reads the water level sensor, controls both servo motors to simulate bridge movement based on water level, and displays relevant information on the LCD. The code is well-commented and easy to understand.	The Arduino code reads the sensor, controls the servo motors, and displays information on the LCD, but there might be minor issues in responsiveness, coordination, or the information displayed. The code has some comments.	The Arduino code attempts to control all components, but there are significant errors in the logic or implementation. The bridge movement and LCD display may not be correctly linked to the water level.	The Arduino code is incomplete, contains major errors, or does not attempt to control all components appropriately.
Integration of Components	The water level sensor, servo motors, and LCD are seamlessly integrated. The system demonstrates a clear and logical relationship between the sensor input, the bridge movement (servo control), and the information displayed on the LCD.	The components are integrated, and a basic relationship between sensor input, servo movement, and LCD display is evident, but there might be areas where the integration could be smoother or more logical.	The integration of the components is weak, and the relationship between the sensor input, servo movement, and LCD display is unclear or inconsistent.	The components are not effectively integrated, and there is little or no clear relationship between them.

Understanding of Concepts	The student demonstrates a clear understanding of how each component works and how they interact within the smart bridge system. They can explain their code and circuit choices effectively.	The student demonstrates a general understanding of the core concepts but might have some minor inaccuracies or struggle to fully explain all aspects of their project.	The student shows a limited understanding of how the components work together or how the Arduino controls them. Their explanations may be unclear or inaccurate.	The student demonstrates a lack of understanding of the fundamental concepts related to the water level sensor, servo motors, LCD, and Arduino control.
Problem Solving	The student effectively troubleshoots any issues that arise during the project and demonstrates resourcefulness in finding solutions.	The student identifies and resolves some issues with the project, possibly with some teacher guidance.	The student struggles to identify and resolve issues with the project and relies heavily on teacher assistance.	The student is unable to identify or resolve problems preventing the project from functioning.
Project Presentation	The student clearly and effectively presents their smart bridge, explaining its functionality and the relationship between the water level, bridge movement,	The student presents their project adequately, explaining its basic functionality.	The student struggles to clearly present their project or explain how it works.	The student is unable to present their project or explain its functionality.

	and LCD information.			
--	----------------------	--	--	--

Now I can... (Self-Assessment):

Students should be asked to reflect on their learning and indicate their level of agreement or understanding with the following statements:

- **Understand the Arduino board and IDE.** (Students should gauge their comfort level with the Arduino hardware and software environment).
- **Design the smart bridge.** (Students should assess their ability to plan and conceptualize the integration of the water level sensor, servo motors, and LCD for the smart bridge).

Teachers can ask students to provide brief explanations for their self-assessment to gain a deeper understanding of their perceived learning.

Relation to Standards:

This project aligns with various educational standards related to science, technology, engineering, art, and mathematics (STEAM), including:

- **Science and Technology Standards:** Understanding sensors, actuators, microcontrollers, data processing, and the application of technology to solve problems.
- **Engineering Design Standards:** Applying the engineering design process to create a solution, including defining the problem, brainstorming, designing, building, testing, and evaluating.
- **Computer Science Standards:** Programming microcontrollers, understanding input and output, control structures, and data display.
- **Mathematics Standards:** Interpreting data from sensors and potentially using mathematical relationships to control the servo motors and LCD display.
- **Digital Literacy:** Effectively using hardware and software tools to create a functional project.

You can use the document of the smart bridge from this link

https://drive.google.com/drive/folders/1xKv4vf7iKGFNiBQJxDrWRNuW_8fdHmxa

